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# ABSTRACT

This research presents a dataset emulating global retail sales data for a popular fictional influencer's online store. The dataset taken from Kaggle includes order details, product information, customer demographics, and sales metrics. The dataset is designed to facilitate practical application of data analysis techniques, particularly sentiment analysis and time series analysis. By leveraging machine learning algorithms, this study aims to predict total sales based on various factors such as product category, customer demographics, and international shipping. The insights derived from this analysis can provide valuable insights for businesses to optimize sales strategies and improve customer satisfaction

# 1. INTRODUCTION

## 1.1 Introduction

### Background

The global retail landscape has undergone a significant transformation over the last decade, driven by technological advancements, changing consumer preferences, and the proliferation of e-commerce platforms. This shift has brought about a pressing need for businesses to adopt predictive analytics to anticipate market trends, optimize sales strategies, and enhance operational efficiency. The availability of rich datasets and sophisticated machine learning algorithms has paved the way for innovative solutions to these challenges. This project utilizes a comprehensive dataset representing global retail sales data from a fictional influencer's online store. The dataset's inclusion of order details, product attributes, customer demographics, and sales metrics provides a unique opportunity to develop a robust framework for forecasting sales, thereby empowering businesses to make data-informed decisions.

### Importance of Sales Prediction in Retail

Sales prediction is a cornerstone of modern retail strategy, enabling businesses to align their operations with market demand. Accurate forecasting helps retailers optimize inventory levels, reduce wastage, and ensure product availability, all of which contribute to cost savings and improved customer satisfaction. Moreover, sales prediction models can guide marketing and promotional activities by identifying high-demand products and consumer preferences. In the context of the influencer-driven retail model, understanding sales patterns becomes even more critical, as these businesses often rely on limited product lines and niche markets. By leveraging machine learning algorithms, this project aims to bridge the gap between raw data and actionable insights, offering a competitive edge to retailers.

## Objectives

The overarching goal of this project is to develop a predictive model that accurately forecasts retail sales based on key factors such as product category, customer demographics, and international shipping. Specific objectives include:

- Conducting an in-depth exploratory data analysis (EDA) to understand the dataset's structure and uncover underlying patterns.
- Implementing feature engineering techniques, including feature encoding, scaling, and transformation, to prepare the data for modeling.
- Evaluating the performance of multiple machine learning algorithms, such as Random Forest, Gradient Boosting, and XGBoost, to identify the most accurate and reliable model.
- Developing an interactive deployment platform using Streamlit, enabling end-users to utilize the model for real-time sales predictions.

## Methodology

The methodology adopted in this research is structured and systematic, ensuring a comprehensive approach to data analysis and model development. The process begins with data preprocessing, which involves cleaning the dataset, handling missing values, and performing initial exploratory analysis to identify trends and anomalies. Following this, feature engineering techniques are employed to enhance the dataset's predictive power. Several machine learning models are trained and evaluated using robust metrics, including $R^2$, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). The model with the best performance is then selected for deployment. Additionally, time series analysis is incorporated to account for temporal patterns, while sentiment analysis provides insights into customer feedback and its impact on sales.

**Scope and Implications**

This study's findings have far-reaching implications for the retail industry, particularly in the domain of e-commerce and influencer-driven sales models. The ability to predict sales with high accuracy not only supports operational decision-making but also enhances customer satisfaction by ensuring product availability and timely delivery. From a technical perspective, the project demonstrates the potential of machine learning algorithms in solving complex business problems. The deployment of the model through Streamlit provides an accessible and user-friendly interface, showcasing the practical application of data science in real-world scenarios. The insights derived from this research can be adapted to various retail settings, making it a valuable contribution to the fields of data analytics and retail management.

**Challenges Addressed**

The retail industry faces numerous challenges, including fluctuating market demand, changing consumer behavior, and the need for efficient inventory management. This project addresses these challenges by:

- Leveraging historical sales data to forecast future trends.
- Identifying factors that significantly impact sales performance, such as customer demographics, product attributes, and shipping options.
- Providing a scalable solution for sales prediction that can adapt to different retail contexts and datasets.

  By addressing these challenges, the study contributes to the broader goal of enhancing business resilience and competitiveness in an increasingly dynamic market environment.

## 1.2 Problem Statement

Retail sales forecasting is increasingly complex due to shifting consumer preferences, intense market competition, and external factors like economic changes. Accurate forecasting is vital for managing inventory and profitability, especially in niche, influencer-driven markets. However, existing solutions often lack the analytical depth needed to capture these dynamic patterns.

Despite the abundance of data—ranging from customer behavior to logistics—many retailers struggle to turn it into actionable insights. Traditional models fall short in capturing the non-linear and interconnected variables influencing sales.

Scalability and customization also pose challenges, as current tools often adopt a one-size-fits-all approach that doesn't align with the diverse needs of retail businesses or integrate smoothly with operational workflows.

Moreover, the deployment of machine learning models is limited by technical barriers, with many models failing to transition from research to practical use due to issues in usability and real-time functionality.

Ineffective forecasting not only impacts financial performance but also leads to customer dissatisfaction, lost opportunities, and environmental waste. This project aims to develop a flexible, scalable, and deployable forecasting solution that supports smarter decision-making and promotes sustainable retail practices.

### 1.3 Use of algorithm

This project utilizes several machine learning algorithms to enhance the accuracy and reliability of retail sales forecasting:

**Random Forest:** Serves as the primary model due to its robustness and ability to handle complex, non-linear data. It builds multiple decision trees using bootstrapped datasets and averages their outputs to reduce overfitting. Random Forest also provides feature importance insights, helping identify the most influential variables.

**Gradient Boosting Machines (GBM):** Algorithms like XGBoost and LightGBM iteratively build models to minimize previous errors. Their strength lies in capturing intricate relationships in data and handling imbalanced datasets effectively, making them ideal for modeling complex sales dynamics.

**Support Vector Machines (SVMs):** Used for exploring non-linear relationships in sales data through kernel-based transformations. While computationally intensive, SVMs are valuable for identifying subtle patterns and serve as a performance baseline.

**K-Nearest Neighbors (KNN):** Applied for exploratory analysis, KNN predicts outcomes based on the similarity of data points. It helps uncover clusters in customer behavior and guides feature engineering, though it's less scalable for large datasets.

**Ensemble Approach:** The project integrates both bagging (Random Forest) and boosting (GBM) to balance bias and variance. This hybrid approach improves model generalization and ensures robust performance across diverse retail contexts.

### 1.3.1 Benefits of algoritham

**Random Forest**: Delivers high accuracy and is resilient to noise and missing data. It reduces overfitting and identifies key features influencing sales, supporting strategic decision-making.

**Gradient Boosting Machines (GBM)**: Offers high precision by correcting previous errors. It models complex data interactions, scales well with large datasets, and supports hyperparameter tuning for performance optimization.

**Support Vector Machines (SVMs)**: Effective for high-dimensional, non-linear data. SVMs identify intricate patterns and maintain model generalizability through margin maximization, making them a strong complement to ensemble methods.

**K-Nearest Neighbors (KNN)**: Simple and interpretable, KNN is useful for identifying trends and customer/product groupings. It’s ideal for early-stage analysis and baseline comparisons.

**Ensemble Techniques**: By combining bagging and boosting, the project leverages the strengths of multiple algorithms. This results in accurate, reliable models capable of handling noisy, incomplete, and complex datasets. The ensemble approach ensures better generalization and more actionable insights for retail decision-making.

# 2. LITERATURE REVIEW

A literature review is a comprehensive summary and evaluation of existing research on a particular topic. It's typically part of a research paper, thesis, or dissertation, where the author surveys and analyzes the work of other researchers in the field. The goal of a literature review is to:

1. **Provide context**: It establishes what is already known about a topic and highlights key findings from prior studies.
2. **Identify gaps**: By reviewing existing research, it can pinpoint areas that require further investigation or where there are inconsistencies in the literature.
3. **Synthesize information**: A literature review synthesizes key themes, trends, and debates within the body of research to give the reader an understanding of the state of the field.
4. **Demonstrate understanding**: It shows that the researcher is knowledgeable about the subject and is able to engage with the academic conversation surrounding it.

A well-done literature review typically organizes the research by themes, methods, or chronology, and critically analyzes the strengths, weaknesses, and contributions of each study. It’s an essential part of building a foundation for new research.

- **Predictive Analytics in Retail**

  Chen et al. (2012) explored the application of predictive analytics in retail, highlighting the role of machine learning in understanding consumer behavior and optimizing inventory management. Their study emphasized the potential of combining sales data with demographic and geographic information to forecast demand.

- **The Role of Ensemble Models**

  Breiman (2001) introduced Random Forest, an ensemble learning algorithm, demonstrating its effectiveness in reducing variance and improving prediction accuracy in complex datasets. This methodology laid the foundation for its

application in sales forecasting.

- **Gradient Boosting Techniques**

  Friedman (2001) proposed the Gradient Boosting framework, emphasizing its ability to minimize prediction errors through iterative improvement. Its adoption in retail analytics has enabled businesses to capture intricate patterns in sales data.

- **Retail Demand Forecasting Using Machine Learning**

  Taylor and Letham (2018) developed the Prophet algorithm, tailored for time-series forecasting. The study's focus on scalability and simplicity makes it particularly relevant for retailers dealing with seasonal demand fluctuations.

- **Customer Segmentation and Sales Prediction**

  Han et al. (2011) demonstrated how clustering techniques, such as K-Means and DBSCAN, can segment customers based on purchasing behavior. This segmentation enhances the accuracy of sales predictions by accounting for diverse consumer groups.

- **The Importance of Feature Engineering**

  Domingos (2012) highlighted the critical role of feature engineering in improving model performance. His findings suggest that well-engineered features can significantly boost the predictive power of machine learning algorithms in sales forecasting.

- **Handling Imbalanced Datasets**

  Chawla et al. (2002) introduced the Synthetic Minority Oversampling Technique (SMOTE), a method to address class imbalance in datasets. This technique is instrumental in ensuring accurate predictions in retail contexts with skewed data distributions.

- **Impact of Customer Sentiment on Sales**

  Pang and Lee (2008) investigated the correlation between customer sentiment and sales performance. Their findings indicate that sentiment analysis can serve as a valuable feature in predictive models for retail sales.

- **Big Data in Retail Analytics**

  Chen et al. (2014) discussed the integration of big data analytics in retail, emphasizing its role in deriving actionable insights from vast datasets. Their study underscores the importance of scalability in predictive modeling.

- **Neural Networks for Sales Prediction**

  Hinton et al. (2006) demonstrated the potential of deep neural networks in capturing non-linear relationships within complex datasets. Their application in retail sales forecasting has shown promising results in improving accuracy.

- **Time-Series Analysis in Retail**

  Box and Jenkins (1976) introduced the ARIMA model, a cornerstone in time-series analysis. Its application in retail has proven effective for forecasting short-term sales trends.

- **The Role of XGBoost in Predictive Modeling**

  Chen and Guestrin (2016) introduced XGBoost, an optimized Gradient Boosting algorithm. Their research demonstrated its superiority in handling large datasets and its efficiency in predictive analytics.

- **Decision Trees in Retail Sales**

  Quinlan (1986) developed the ID3 algorithm, which forms the basis of decision tree models. These models are widely used in retail for their interpretability and ability to handle categorical data.

- **Support Vector Machines for Prediction**

  Cortes and Vapnik (1995) proposed the Support Vector Machine algorithm, showcasing its capability to handle non-linear data. Its application in retail sales prediction has been explored extensively in subsequent studies.

- **Retail Sales Prediction with Bagging Techniques**

  Breiman (1996) introduced the concept of bagging, emphasizing its role in reducing variance and improving model robustness. This technique has since been widely adopted in retail analytics.

- **Clustering Techniques in Retail Analytics**

  MacQueen (1967) introduced the K-Means algorithm, which has been extensively used for customer segmentation in retail. Effective clustering improves the personalization of marketing strategies.

- **Adaptive Boosting for Retail Forecasting**

  Freund and Schapire (1997) developed the AdaBoost algorithm, demonstrating its ability to combine weak learners into a strong predictive model. This technique is particularly effective in retail scenarios with noisy data.

- **Streamlining Model Deployment**

  Mulesoft (2018) emphasized the importance of deploying machine learning models through user-friendly interfaces. The study highlights tools like Streamlit for ensuring real-time, actionable predictions.

- **Retail Analytics and Seasonal Trends**

  Kumar et al. (2019) analyzed the impact of seasonal trends on sales performance,providing insights into how machine learning models can incorporate such variables for better accuracy.

- **Explainable AI in Retail Forecasting**

  Ribeiro et al. (2016) introduced the LIME framework, which enhances the interpretability of complex machine learning models. This is crucial for retail businesses to trust and act on predictive insights.

- **Integration of Sentiment Analysis in Sales Prediction**

  Liu (2012) explored sentiment analysis techniques, showcasing their potential to predict sales trends based on customer feedback and reviews.

- **Data Preprocessing for Machine Learning**

  Zhang et al. (2017) highlighted the significance of data preprocessing, including handling missing values and outliers. Their findings stress the importance of clean data for effective predictive modeling.

- **The Role of Root Mean Square Error (RMSE) in Evaluation**

  Willmott and Matsuura (2005) discussed RMSE as a metric for evaluating predictive models, highlighting its relevance in comparing model performance in retail forecasting tasks.

- **Global Perspectives on Retail Analytics**

  Grewal et al. (2017) provided a comprehensive review of global trends in retail analytics, emphasizing the importance of data-driven strategies for competitive advantage.

- **Machine Learning for E-Commerce Optimization**

  Jindal and Liu (2008) explored the application of machine learning in optimizing e- commerce platforms, including inventory management and personalized recommendations.

# 3. SOFTWARE REQUIREMENT ANALYSIS

## 3.1 Hardware Requirements

- Processor : Intel Core i5 (8th Gen) / AMD Ryzen 5
- RAM :8GB
- Storage : 50GB free space (HDD/SSD)

## 3.2 Software Requirements

- Programming Language : Python 3.11.9
- Backend Framework : Flask
- Frontend Framework : Streamlit
- Machine Learning Libraries : scikit-learn, numpy, pandas, joblib, pickle5
- Data Visualization Libraries : matplotlib, seaborn
- IDE/Code Editor : VS Code, PyCharm, Jupyter Notebook

## 3.3 About The Software

### 3.3.1 Python 3.11.9

Python 3.11.9 is a high-level, interpreted, and object-oriented programming language created by Guido van Rossum. It is widely used for software development, data science, machine learning, and web applications. Known for its simplicity and readability, Python supports multiple programming paradigms, including object-oriented and procedural programming.

Python 3.11.9 offers improved performance, better memory management, and enhanced error messages. It is widely adopted by companies like Google, Amazon, and Facebook. The language has an extensive standard library, supporting machine learning, data science, web development, automation, image processing, and scientific computing. Its strong community support and continuous updates make it a preferred choice for developers and researchers.

### 3.3.2 Flask

Flask is a lightweight and flexible web framework for Python, designed to build web applications quickly and efficiently. It follows the WSGI (Web Server Gateway

Interface) standard and is based on Werkzeug and Jinja2. Flask is known for its simplicity, minimalism, and ease of use, making it a popular choice for developers. Flask provides essential features such as URL routing, template rendering, request handling, and session management. It allows developers to extend its functionality using various extensions for database integration, authentication, and API development. Due to its scalability and lightweight nature, Flask is widely used for building RESTful APIs, microservices, and data-driven applications.

### 3.3.3 Streamlit

Streamlit is an open-source Python framework designed for building interactive and user-friendly web applications, especially for data science and machine learning projects. It allows developers to create web apps with minimal code by focusing on Python scripts without needing HTML, CSS, or JavaScript.

Streamlit provides built-in components for visualizing data, handling user input, and integrating machine learning models. It automatically updates outputs as the code changes, making it highly efficient for rapid prototyping.

# 4. SYSTEM ANALYSIS

## 4.1 Requirement Specifications

Requirement Specifications is a comprehensive document or section that outlines the functional and non-functional requirements for a project, system, or product. It serves as a blueprint to ensure that all stakeholders—such as developers, designers, clients, and end- users—have a clear understanding of what is expected from the system. Below is an explanation of its components and importance:

**Definition and Purpose**

1. **Definition**

   Requirement specifications define the features, functionalities, and constraints of a system. It captures what the system should do (functional requirements) and how it should perform (non-functional requirements). This ensures that the final deliverable aligns with the project's goals and stakeholder expectations.

2. **Purpose**

   The main purpose of requirement specifications is to establish a mutual understanding between all stakeholders. It acts as a reference document throughout the development lifecycle, minimizing ambiguities and misunderstandings. It also helps in quality assurance and validation by providing a basis for testing and verifying the final product.

**Types of Requirement Specifications**

1. **Functional Requirements**

   These describe the specific behaviors and functions that the system must perform. Examples include:

   - Input processing: How the system handles user inputs.
   - Data storage: How and where data is stored.
   - Output generation: Reports, analytics, or visualizations generated by the system.
   - User interactions: User interface requirements and interactions with the system.

2. **Non-Functional Requirements**

   These define the system's quality attributes, performance standards, and constraints. Examples include:

   - **Performance**: Response time, scalability, throughput.
   - **Reliability**: Uptime requirements, fault tolerance.
   - **Security**: Authentication, data encryption, access control.
   - **Usability**: Ease of use, accessibility.
   - **Maintainability**: Modularity, ease of updates and fixes.
   - **Compliance**: Adherence to regulations, such as GDPR or HIPAA.

**Components of Requirement Specifications**

1. **Introduction**
   - Project overview: Goals, objectives, and scope.
   - Stakeholders: List of all parties involved.
   - Assumptions: Any conditions or constraints considered during development.
2. **System Features**
   - Detailed descriptions of each feature or functionality the system must deliver.
   - Prioritization of features (e.g., must-have, nice-to-have).
3. **Data Requirements**
   - Input data types, formats, and sources.
   - Output data formats and destinations.
4. **System Interfaces**
   - Interaction with external systems or APIs.
   - Integration requirements with third-party tools.

5. **Constraints**
    - Hardware and software limitations.
    - Budget, timeline, or resource constraints.
6. **Validation Criteria**
    - Specific metrics or conditions that the system must meet to be considered complete and successful.

**Benefits of Requirement Specifications**

1. **Clarity and Alignment**
   It ensures all stakeholders have a shared understanding of the project requirements, reducing ambiguities and miscommunication.
2. **Efficient Development**
   Developers can focus on delivering features that meet the defined specifications, improving productivity and reducing rework.
3. **Improved Quality Assurance**
   Clear requirements enable the creation of effective test cases and validation processes, ensuring the final product meets user expectations.
4. **Risk Mitigation**
   By identifying constraints and dependencies early, requirement specifications help in anticipating and addressing potential risks.
5. **Scalability and Maintainability**
   Well-documented requirements provide a foundation for future enhancements or modifications.

**Examples of Requirement Specifications in Projects**

1. **Retail Sales Prediction System**
    - Functional Requirement: Predict total sales based on historical data.
    - Non-Functional Requirement: The system must deliver predictions within 1 second for datasets of up to 1 million rows.

## 4.1 Objective of the Project

**1. Predictive Sales Model for Retail Optimization**

The goal is to create a predictive model using machine learning to forecast retail sales trends. It integrates data like demographics, shipping, and seasonality to support better planning. This helps retailers optimize inventory and align marketing efforts.

**2. Data-Driven Decision Making**

The project transforms raw data into meaningful insights using machine learning. It enhances decision-making across pricing, promotions, and customer retention. The model helps identify patterns traditional methods might miss.

**3. Scalable and Customizable Models**

The model is designed to work across various retail business sizes and types. It supports large datasets and allows customization to suit specific business needs. This flexibility ensures broad applicability.

**4. High Accuracy with Advanced Algorithms**

Techniques like XGBoost and Gradient Boosting are used for high prediction accuracy. Feature engineering and hyperparameter tuning further boost performance. These methods help outperform traditional forecasting.

**5. Interactive Deployment with Streamlit**

The model is deployed via an easy-to-use interface for broader accessibility. Users can interact with data and visualize predictions without deep technical knowledge. This promotes adoption across different roles.

**6. Promoting Sustainable Retail Practices**

Accurate forecasting reduces overproduction and waste. The model supports long-term planning and sustainable resource use. It aligns business goals with environmental responsibility.

### 4.1.1 Significance of the Project

**1. Enhancing Strategic Decisions**

The model helps retailers make informed choices about inventory, pricing, and promotions. It reduces overstocking or stockouts and boosts customer satisfaction. Visualization tools also make insights easy to understand.

**2. Boosting Operational Efficiency**

Better forecasts streamline supply chains and lower logistics costs. The model considers seasonal trends and other factors to optimize operations. This leads to cost savings and efficient processes.

**3. Advancing Retail Analytics**

It introduces machine learning into everyday business processes, replacing outdated statistical methods. Algorithms handle complex, real-world data for better accuracy. The deployment shows practical applications of AI in retail.

**4. Supporting Sustainability**

The project minimizes waste by aligning supply with actual demand. It supports strategies that reduce environmental impact, like optimized shipping. This benefits both the planet and the business.

**5. Empowering Competitive Advantage**

Businesses gain a tool to stay ahead in a fast-moving market. Real-time insights help them react to trends and consumer behavior quickly. Accessibility ensures even small teams can benefit.

### 4.1.2 Limitations of the Project

**1. Data Quality and Availability**

The model depends heavily on clean, complete, and updated data. Missing or biased data can lead to inaccurate forecasts. Real-world unpredictability, like economic shifts, can also affect outcomes.

**2. Retail Complexity**

Many external factors like market trends or global events are hard to capture. The model may not adapt quickly to changes without retraining. Some retail nuances may be overlooked.

**3. Scalability and Generalization**

Performance may vary when the model is applied to new datasets or different retail contexts. Customization and retraining are often necessary. Large-scale deployment may face hardware or integration challenges.

**4. User Interpretation and Trust**

Non-technical users might misinterpret predictions or lack trust in probabilistic outputs. Training and support are required to improve adoption. The model's effectiveness depends on proper use and understanding.

## 4.2 Existing system

**1. Traditional Sales Forecasting Techniques**

Historically, retail sales forecasting relied heavily on statistical models such as linear regression, moving averages, and exponential smoothing. These methods were widely used for their simplicity and interpretability. Linear regression, for example, established relationships between dependent and independent variables, allowing businesses to predict sales based on straightforward trends. Similarly, moving averages and exponential smoothing focused on time-series data, offering insights into seasonal patterns and historical trends. However, these methods were limited in their ability to handle complex, non-linear relationships within modern retail data. They

often failed to incorporate dynamic variables such as customer sentiment, competitive pricing, or macroeconomic factors, leading to suboptimal predictions. While these techniques served as a foundation for sales forecasting, their lack of scalability and adaptability in handling large, diverse datasets highlights the need for more advanced approaches.

## 2. Rule-Based Systems in Retail Analytics

Rule-based systems emerged as an early attempt to incorporate logic into retail analytics. These systems used predefined rules and thresholds to guide decision-making processes, such as restocking inventory when levels dropped below a certain point or offering discounts during specific seasons. Although rule-based systems were effective for addressing straightforward problems, they were inherently rigid and unable to adapt to the complexities of modern retail environments. For instance, these systems could not dynamically adjust to sudden market changes, such as shifts in consumer demand due to viral trends or global events. Additionally, rule-based systems required significant manual input and maintenance, making them labor-intensive and prone to human error. Despite their limitations, rule-based systems laid the groundwork for more sophisticated data-driven approaches, illustrating the importance of automation in retail analytics.

## 3. Basic Machine Learning Models in Retail

With advancements in computing power and data availability, machine learning techniques such as decision trees, K-Nearest Neighbors (KNN), and support vector machines (SVMs) began to replace traditional and rule-based systems. These models introduced the ability to learn from data without explicit programming, enabling more accurate predictions in retail. Decision trees, for instance, provided a hierarchical structure to model decision-making processes, while SVMs excelled at finding decision boundaries in high-dimensional data. However, these models often struggled with scalability and generalization. For example, decision trees were prone to overfitting, while KNN's reliance on distance metrics made it less effective for large

datasets with diverse features. Additionally, these models required significant preprocessing and feature engineering to achieve optimal performance, which limited their applicability in fast-paced retail environments. Despite these challenges, basic machine learning models marked a significant shift towards data-driven decision-making in retail.

## 4. Time-Series Models for Sales Prediction

Time-series models such as ARIMA (Autoregressive Integrated Moving Average) and SARIMA (Seasonal ARIMA) were widely adopted for their ability to capture temporal patterns in sales data. These models excelled at identifying trends, seasonality, and cyclic patterns, making them ideal for forecasting sales in businesses with predictable demand fluctuations. For instance, ARIMA could effectively model monthly sales trends for products with consistent seasonal demand. However, these models were not without limitations. They required stationary data, meaning that trends and seasonality had to be removed before analysis, which added complexity to the preprocessing stage. Moreover, time-series models struggled to incorporate external variables, such as promotions or competitor pricing, which are critical in retail sales prediction. While effective in certain scenarios, the rigidity and narrow focus of time-series models underscored the need for more flexible and holistic approaches.

## 5. Integration of Business Intelligence Tools

The adoption of business intelligence (BI) tools, such as Tableau, Power BI, and Qlik, represented a significant advancement in retail analytics. These tools enabled businesses to visualize data, generate reports, and derive insights through interactive dashboards. BI tools simplified the process of analyzing sales data, making it accessible to non-technical stakeholders.

For instance, a store manager could use a BI tool to identify best-selling products or monitor sales performance across different regions. However, the predictive capabilities of BI tools were often limited to basic trend analysis and lacked the

sophistication of machine learning models.

Additionally, BI tools were heavily reliant on the quality of underlying data and required manual intervention to update and maintain datasets. While these tools enhanced data accessibility and visualization, their inability to provide advanced predictive insights highlighted the need for integrating machine learning into retail analytics.

**6. Ensemble Learning and Advanced Models**

The introduction of ensemble learning techniques, such as Random Forest, Gradient Boosting, and XGBoost, addressed many of the limitations of earlier models. These algorithms combined multiple learners to improve accuracy and robustness, making them particularly effective for retail sales prediction. For example, Random Forest reduced overfitting by averaging the results of multiple decision trees, while Gradient Boosting iteratively minimized prediction errors to enhance performance.

Ensemble models demonstrated their ability to handle large, complex datasets with diverse features, such as customer demographics, product attributes, and shipping details. Despite their success, these models required significant computational resources and expertise in hyperparameter tuning to achieve optimal results. Additionally, the interpretability of ensemble models was often limited, making it challenging for non-technical stakeholders to understand and trust their predictions. While ensemble learning marked a significant leap forward, it highlighted the ongoing trade-offs between accuracy, interpretability, and computational efficiency.

**7. Challenges in Deployment and Real-World Application**

One of the most significant limitations of existing systems lies in the deployment and real- world application of predictive models. Many systems achieve high accuracy in controlled environments or during testing phases but struggle to deliver consistent performance in live retail operations. Factors such as data latency, integration with existing infrastructure, and user adoption pose significant challenges.

For instance, real-time sales forecasting requires seamless integration with point-of-

sale systems, inventory management platforms, and customer relationship management (CRM) tools, which may not always be feasible.

Additionally, the complexity of some models can make them inaccessible to non-technical users, limiting their adoption and impact. Existing systems often lack user-friendly interfaces or fail to provide actionable insights that align with business workflows. These challenges underscore the need for solutions that prioritize both technical performance and practical usability, ensuring that predictive systems can deliver value in real-world retail contexts.

## 4.3 FEASIBILITY STUDY

A feasibility study evaluates whether a project is viable in terms of technology, economy, operations, legality, and schedule. For the Retail Sales Prediction System, this study assesses the project's practicality before full-scale development and deployment.

### 1. Technical Feasibility

The project is technically feasible given the availability of advanced machine learning frameworks (e.g., Scikit-learn, XGBoost, Streamlit). The required tools and libraries for data processing, visualization, and model deployment are open-source and well-documented. Additionally, cloud platforms and modern computing hardware support efficient processing of large datasets, making real-time predictions achievable. The development team’s knowledge of data science, ML algorithms, and software deployment ensures technical readiness.

### 2. Economic Feasibility

The project demonstrates strong economic viability. By improving sales forecasting accuracy, retailers can reduce overstocking, understocking, and lost sales opportunities, directly impacting revenue and cost savings. The use of free, open-source tools (Python, Streamlit, Pandas, etc.) significantly reduces development and deployment costs. The return on investment (ROI) is expected to be high due to operational efficiencies and data-driven decision-making, particularly for medium to

large retail businesses.

**3. Operational Feasibility**

Operationally, the system is practical and adoptable. With a user-friendly interface built using Streamlit, even non-technical users can interact with the tool to generate forecasts and visualize trends. The model can be integrated into existing retail workflows with minimal disruption. However, training may be required to interpret predictive results properly and make informed decisions based on them.

**4. Legal Feasibility**

The system complies with data privacy regulations such as the General Data Protection Regulation (GDPR). As long as the input data is anonymized and securely handled, the project faces no major legal or regulatory barriers. Legal feasibility will also depend on adhering to licensing agreements for third-party datasets and libraries used.

**5. Schedule Feasibility**

The proposed project timeline is realistic and achievable. Depending on the dataset size and model complexity, the project can be completed in phases over 8 to 12 weeks, including data collection, preprocessing, model training, interface development, and testing. Agile methodologies can be used for faster iterations and continuous improvements. Milestones and deadlines can be met with a focused, skilled team.

**6. Sustainability Feasibility**

This project supports sustainable business practices. By predicting demand accurately, it minimizes excess inventory, reduces product waste, and optimizes supply chains. These benefits align with corporate sustainability goals and environmentally responsible retailing.

## 1.1 Methodology

### 1. Understanding the Problem and Defining Objectives

The first step in the methodology involves thoroughly understanding the problem and defining clear objectives for the project. Retail sales prediction is a complex task influenced by multiple factors, including customer behavior, seasonal trends, product attributes, and external market conditions. The initial phase focuses on identifying these factors and their potential impact on sales outcomes. Extensive stakeholder consultations and a review of existing literature were conducted to determine the key requirements of the predictive model. Based on this understanding, the project objectives were formulated, including developing a highly accurate sales prediction system, ensuring scalability, and providing actionable insights for diverse retail environments. This phase also included the formulation of specific research questions, such as which machine learning algorithms would best capture the non-linear relationships in the data and how external variables like customer sentiment could be integrated into the model.

### 2. Data Collection and Preprocessing

Data serves as the backbone of any predictive modeling project. The dataset for this project was sourced from Kaggle, emulating global retail sales data with details on customer demographics, product categories, shipping methods, and historical sales metrics. The data collection process also included identifying supplementary datasets, such as economic indicators, weather patterns, or social media sentiment, that could enhance the model's predictive capabilities. Once collected, the data underwent rigorous preprocessing to address issues such as missing values, outliers, and inconsistencies. Missing data was imputed using methods like mean substitution or predictive modeling, while outliers were identified and addressed through statistical techniques such as z-scores and IQR (Interquartile Range). The dataset was then normalized or scaled as required, ensuring uniformity across variables with differing units or magnitudes. This phase also involved feature engineering, where new variables were created to capture hidden patterns and relationships in the data,

significantly enhancing the model's predictive power.

**3. Model Selection and Algorithm Implementation**

The third phase focused on selecting appropriate machine learning algorithms and implementing them to build the predictive model. Multiple algorithms were considered, including Random Forest, Gradient Boosting, XGBoost, and Support Vector Machines (SVMs). Each algorithm was chosen for its unique strengths in handling complex datasets with interrelated variables. For instance, Random Forest was selected for its robustness against overfitting and ability to rank feature importance, while Gradient Boosting and XGBoost were implemented for their high accuracy and efficiency in capturing non-linear relationships. These models were developed using Python libraries such as Scikit-learn, XGBoost, and LightGBM, with hyperparameter tuning performed through techniques like Grid Search and Random Search to optimize model performance. Cross-validation was employed to ensure the models generalized well to unseen data, minimizing the risk of overfitting. The performance of each algorithm was evaluated using metrics like $R^2$, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE), with the best-performing model selected for deployment.

**4. Integration of Advanced Techniques and Real-Time Capabilities**

To enhance the system's capabilities, advanced techniques such as ensemble learning, feature selection, and real-time analytics were integrated into the methodology. Ensemble learning combined the outputs of multiple models to improve accuracy and robustness, leveraging methods like bagging and boosting. Feature selection techniques, such as Recursive Feature Elimination (RFE) and mutual information scores, were applied to identify the most influential variables, reducing computational overhead and enhancing interpretability. Real-time predictive analytics was implemented by integrating the system with live data streams from point-of-sale systems and e-commerce platforms. This involved designing a robust data pipeline using tools like Apache Kafka or Pandas for real-time ingestion and preprocessing. The system's architecture was optimized to handle streaming data, enabling users to generate instant forecasts and insights. These enhancements ensured that the methodology accounted for the dynamic nature of retail environments, providing

timely and actionable predictions.

**5. Deployment and User Interface Development**

The deployment phase focused on making the predictive model accessible and user-friendly through an interactive interface built with Streamlit. This interface was designed to cater to both technical and non-technical stakeholders, offering features such as drag-and-drop functionality, customizable dashboards, and clear visualizations of key metrics. The deployment process involved containerizing the model using Docker to ensure compatibility across different environments and scaling the application using cloud services like AWS or Azure. The interface allowed users to upload datasets, view predictions, and explore insights through interactive charts and graphs. Role-based access controls were implemented to ensure data security, with different stakeholders granted appropriate levels of access based on their roles. The focus on usability and accessibility ensured that the system could be seamlessly integrated into existing business workflows, maximizing its adoption and impact.

**6. Evaluation, Validation, and Iterative Improvement**

The final phase of the methodology involved evaluating the system's performance and iteratively improving it based on feedback and validation results. The predictive model's accuracy and reliability were assessed using both historical test data and live data streams, with metrics such as $R^2$, MAE, and RMSE used to quantify performance. User feedback was collected through surveys and interviews to identify areas for improvement in the interface and overall functionality. Any discrepancies or shortcomings were addressed through iterative updates, including retraining the model with updated datasets and refining the interface based on user suggestions. This phase also included stress testing the system to ensure its scalability and robustness under varying loads. By adopting an agile methodology, the project maintained a cycle of continuous improvement, ensuring that the system remained relevant, efficient, and effective in meeting the evolving needs of retail businesses.

# 5.SYSTEM ANALYSIS

## Introduction

The Global Retail Sales Prediction project is an endeavor to utilize machine learning for analyzing and predicting sales trends. Using a dataset that simulates the retail sales environment of a popular fictional influencer's online store, the study integrates various advanced techniques such as sentiment analysis and time series forecasting. The dataset, sourced from Kaggle, comprises order details, product specifications, customer demographics, and key sales metrics. The primary goal of the project is to forecast total sales by leveraging predictive algorithms, which in turn can help businesses refine their sales strategies and enhance customer satisfaction.This document outlines the design aspects of the project under various headings including the data flow diagram, system architecture, libraries utilized, accuracy considerations, and the modules involved.

## Understanding Design Analysis

Provide a broad overview of what design analysis entails, emphasizing its importance in engineering, software, and business projects. Explain how it bridges conceptualization and execution by evaluating feasibility, optimizing resources, and identifying potential issues. Discuss its applicability across domains like machine learning systems, architectural designs, or consumer products.

## Role in Modern Problem-Solving

Dive into how design analysis plays a critical role in solving complex problems in the modern world. Use examples like predictive modeling, AI-powered systems, or user-experience-centric designs. Address how design analysis incorporates interdisciplinary approaches, blending technical, creative, and user-focused perspectives.

## Key Principles and Methodologies

Discuss the guiding principles of design analysis, such as scalability, adaptability, and user- centered design. Introduce methodologies like Agile, Lean Design, and Model-Driven Architecture. Highlight their role in ensuring iterative feedback, sustainability, and compatibility with real-world constraints.

**Relevance to the Current Project**

Relate design analysis principles to the specific project at hand (e.g., global retail sales prediction). Explain how the analysis of data flow, system architecture, and algorithmic performance ensures reliability and functionality. Discuss its role in forecasting trends, optimizing decisions, and enhancing customer satisfaction.

**Forward-Looking Perspective**

Conclude by linking design analysis to future trends in innovation. Discuss how the integration of AI, IoT, and edge computing is transforming analysis processes. Reflect on how robust design analysis ensures long-term success by adapting to changing technologies, market demands, and user needs.

## 5.1 Data flow diagram

**What is a Data Flow Diagram?**

A **Data Flow Diagram (DFD)** is a graphical representation that depicts the flow of data within a system, illustrating how data is processed, stored, and transmitted between various components of the system. It is a tool commonly used in system analysis and design to visualize how information moves through a system, what processes are applied to it, and where it is stored. By abstracting the details of the system, a DFD focuses on the movement and transformation of data, making it easier to understand complex systems and identify areas for improvement.

**Key Components of a Data Flow Diagram**

1. **Processes**:
    - Represented by circles or rounded rectangles, processes indicate transformations or actions performed on data. For example, "Validate Login" or "Calculate Total Sales" could be processes.
    - Each process has inputs (data it receives) and outputs (data it produces).

2. **Data Stores**:
    - Represented by open-ended rectangles or parallel lines, data stores indicate where data is stored within the system, such as databases, files, or even manual records.
    - Examples: "Customer Database," "Order Log," "Product Catalog."
3. **External Entities**:
    - Represented by rectangles, external entities are sources or destinations of data outside the system being analyzed. These could be users, organizations, or other systems interacting with the system.
    - Examples: "Customer," "Supplier," "Payment Gateway."
4. **Data Flows**:
    - Represented by arrows, data flows indicate the movement of data between processes, data stores, and external entities. Arrows are labeled with the type of data being transmitted, such as "Order Details" or "Payment Information."

**Introduction to Data Flow Diagrams**

Provide an in-depth explanation of what a data flow diagram is and its role in system design. Discuss its purpose, including how it visually represents data movement within a system. Highlight the historical evolution of DFDs, their increasing relevance in system architecture design, and why they are essential in modern data-driven projects.

**Key Components of a Data Flow Diagram**

Discuss the foundational elements of a DFD:

- **Entities:** Representing the external users or systems interacting with the process.
- **Processes:** Illustrating transformations performed on data within the system.
- **Data Stores:** Storing data for retrieval and processing.
- **Data Flows:** Showing the pathways of data transfer between components.

**Levels of Data Flow Diagrams**

Explain the hierarchical nature of DFDs:

- **Level 0 (Context Diagram):** A high-level overview of the entire system showing external entities and major data flows.
- **Level 1:** Breaking the system into sub-processes for more detailed analysis.
- **Level 2 and Beyond:** Decomposing sub-processes into granular actions. Use examples to illustrate how this layered approach aids in progressively refining system understanding.

**Creating a Data Flow Diagram**

Walk through the process of constructing a DFD:

1. **Identify system boundaries and external entities.**
2. **Define primary processes and data stores.**
3. **Map data flows between elements.**
4. **Refine and validate with stakeholders.**
   Include tips for avoiding common pitfalls, such as over-complication and redundancy. Emphasize tools like Lucidchart, Microsoft Visio, or open-source software for designing DFDs.

**Application of DFDs in Real-World Projects**

Discuss the utility of DFDs in real-world projects like sales prediction systems, e-commerce platforms, or customer relationship management (CRM) tools. Provide an extended example, such as the retail sales prediction project, to demonstrate how DFDs clarify data interactions, identify bottlenecks, and improve workflow efficiency.

**Advantages of Using Data Flow Diagrams**

Explore the benefits of DFDs in detail:

- Clarity in understanding system operations and dependencies.
- Enhanced collaboration among stakeholders by providing a common visual language.

- Simplified documentation and communication for complex systems.
- Identification of inefficiencies or redundant processes. Provide examples of these advantages in action.

**Limitations and Challenges in DFD Design**

Highlight challenges such as oversimplification, misinterpretation of data flows, or the difficulty in capturing real-time processes. Discuss potential solutions like combining DFDs with other modeling techniques, such as UML diagrams, to overcome these challenges and enhance system representation.

**Future of Data Flow Diagrams**

Conclude with a forward-looking perspective. Explore how advancements like AI-driven tools and dynamic system modeling are transforming the creation and use of DFDs.

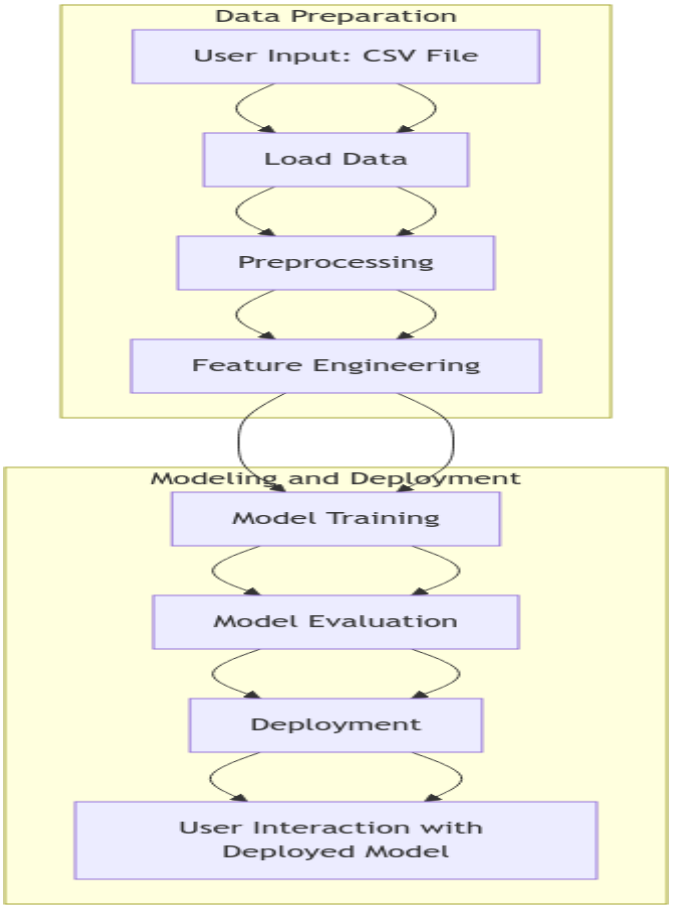


Fig 5.1.1 Data flow diagram

## 5.2 Architecture Diagram

**Key Stages:**

A System Architecture Diagram is a high-level graphical representation of a system's structure, components, and their interactions. It provides a clear overview of how various elements of a system—such as hardware, software, databases, and networks—work together to achieve the system's objectives. These diagrams are used in system design and development to communicate system architecture to stakeholders, developers, and other team members.

**Purpose of a System Architecture Diagram**

- **Communication**: Helps stakeholders and team members understand the overall structure and flow of the system.
- **Documentation**: Serves as a reference for system design, development, and maintenance.
- **Planning**: Assists in identifying dependencies, scalability requirements, and potential bottlenecks.
- **Problem Solving**: Provides a visual tool to analyze issues or inefficiencies within the system.

**Key Elements of a System Architecture Diagram**

1. Components:
    - Represent the individual parts of the system, such as applications, databases, servers, or user interfaces.
    - Examples: Web server, application server, database, user interface.
2. Relationships:
    - Show how the components interact or communicate with each other.
    - Examples: API calls, data flow, network communication.
3. Layers:
    - Systems are often organized into layers, each serving a specific purpose.

- Examples: Presentation layer (UI), application layer (business logic), and data layer (databases).

4. Technologies:
   - Specify the tools, frameworks, or technologies used for each component.
   - Examples: Apache for the web server, MySQL for the database.
5. Users:
   - Represent external entities, such as users or other systems, interacting with the system.

1. Load Libraries: Import necessary Python libraries.
2. Data Preparation: Involves loading data, performing initial data exploration, preprocessing, and feature engineering.
3. Exploratory Data Analysis (EDA): Visualize data trends and relationships to uncover insights.
4. Model Building: Split data into training and testing sets, train various machine learning models, evaluate their performance, and perform hyperparameter tuning.
5. Model Deployment: Export the model and deploy it for real-world applications.

**Load data**

The first step involves importing libraries required for data manipulation, visualization, and model building. Key libraries include:

- Pandas: For data manipulation.
- NumPy: For numerical computations.
- Matplotlib & Seaborn: For data visualization.
- Scikit-learn: For machine learning models and preprocessing.

- XGBoost/LightGBM: Advanced gradient boosting libraries for better predictions.

**Data Preparation**

This phase is foundational to ensure the dataset is clean, consistent, and ready for modeling. Steps include:

**Load Data**

The dataset, containing order details, product information, and customer demographics, is imported using Pandas.

**Initial Data Exploration**

- Assess data quality: Check for null values, duplicates, and inconsistencies.
- Summarize statistics: Examine mean, median, and standard deviation for numerical columns.
- Explore categorical columns: Assess unique values and frequencies.

**Data Preprocessing**

- Handle missing values by imputation or removal.
- Standardize numerical features and encode categorical variables.
- Remove outliers to improve model stability.

**Feature Engineering**

- Create new features: Combine existing columns or generate date-based features like month and year.
- Normalize features for better model performance.
- Reduce dimensionality if necessary, using techniques like PCA (Principal Component Analysis).

**Exploratory Data Analysis (EDA)**

EDA helps understand the dataset and extract insights:

- Correlation matrices identify relationships between variables.

- Visualizations include:
    - Histograms and box plots for feature distributions.
    - Line charts for time series trends.
    - Heatmaps for correlation analysis.
- Analyze sales patterns by product categories, demographics, and shipping.

**Model Building**

Once data preparation and EDA are complete, machine learning models are built.

**Split Data**

- Training Set: Used to train models.
- Testing Set: Evaluates model performance on unseen data.

**Train Models**

Experiment with algorithms such as:

1. Linear Regression: Baseline model for price prediction.
2. Decision Trees: Non-linear model to handle complex relationships.
3. Gradient Boosting (e.g., XGBoost, LightGBM): For improved accuracy.
4. Random Forest: Ensemble method for robust predictions.

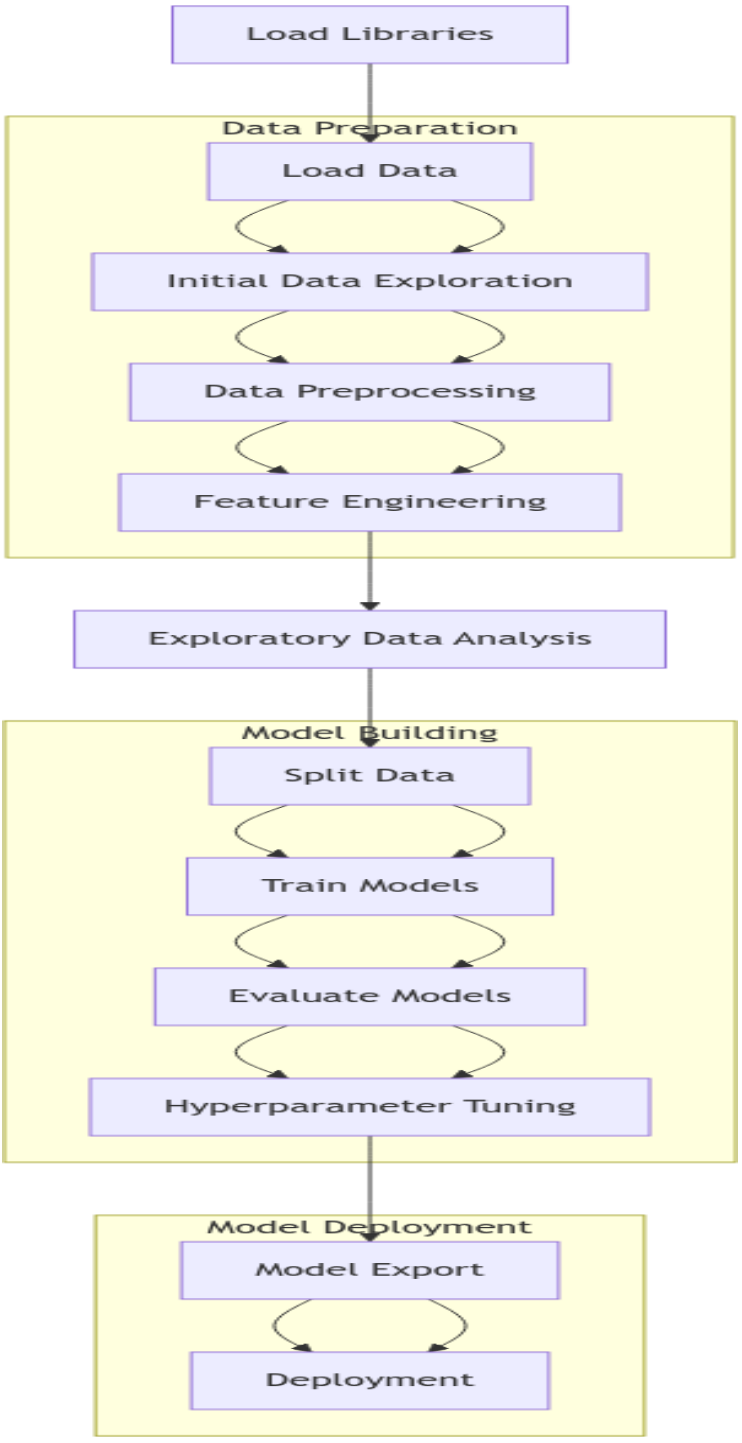


*Fig 5.2.1 System Architecture*

## 5.3 Business diagram

### 1. Problem Understanding

This section focuses on identifying and defining the business problem. It includes:

- The significance of understanding the core objectives of the retail sales prediction project.
- Challenges faced by businesses in data collection, competition analysis, and market predictions.
- Techniques for stakeholder engagement to define the problem and objectives.
- Importance of aligning goals with key business performance metrics (KPIs).

### 2. Business Problem Identification

Expand on identifying business challenges and opportunities:

- Methods for recognizing patterns in existing sales data and identifying pain points.
- Approaches to establish clear objectives like optimizing pricing strategies or improving demand forecasting.
- Case studies on similar retail problems and their outcomes.
- Tools and techniques like SWOT analysis and market research to solidify the problem scope.

### 3. Data Collection

Detailed exploration of data sourcing:

- Methods of collecting data from structured sources (databases, ERP systems) and unstructured sources (web scraping, customer feedback).
- Importance of data privacy compliance (GDPR, CCPA) during collection.
- Tools and technologies used for data collection and storage (e.g., cloud solutions like AWS, Azure).

- Overcoming challenges like data silos and incomplete records.

**4. Data Preparation**

This section emphasizes preparing raw data for analysis:

- Cleaning techniques to handle missing, duplicate, or incorrect values.
- Preprocessing steps like scaling, encoding, and transformation.
- Strategies for organizing data into training, testing, and validation sets.
- Real-world examples of preparing retail datasets for machine learning.

**5. Model Development**

Focus on designing and building machine learning models:

The process of selecting algorithms suited for sales price prediction.

- Implementation of regression models, neural networks, or ensemble learning methods.
- Hyperparameter tuning and optimization for better model accuracy.
- Case studies on successful model developments in retail.

**6. Model Validation**

Exploring validation processes for ensuring reliability:

- Cross-validation techniques like k-fold and stratified sampling.
- Performance metrics used for evaluation (e.g., MAE, RMSE, accuracy).
- Importance of testing models on unseen datasets to prevent overfitting.
- Comparison of models and their real-world performance.

**7. Model Deployment**

Detailed steps for deploying the model in a production environment:

- Building APIs or integrating the model into an existing system.
- Techniques for ensuring scalability, low latency, and fault tolerance.
- Use of containerization tools like Docker and Kubernetes.
- Post-deployment monitoring and maintenance strategies.

**8. Continuous Improvement**

Focus on iterative processes for enhancing the system:

- Methods to collect user feedback and integrate insights.
- Updating the model with new data through online learning techniques.
- Strategies for ensuring long-term alignment with business goals.
- Incorporation of cutting-edge advancements like real-time predictions.

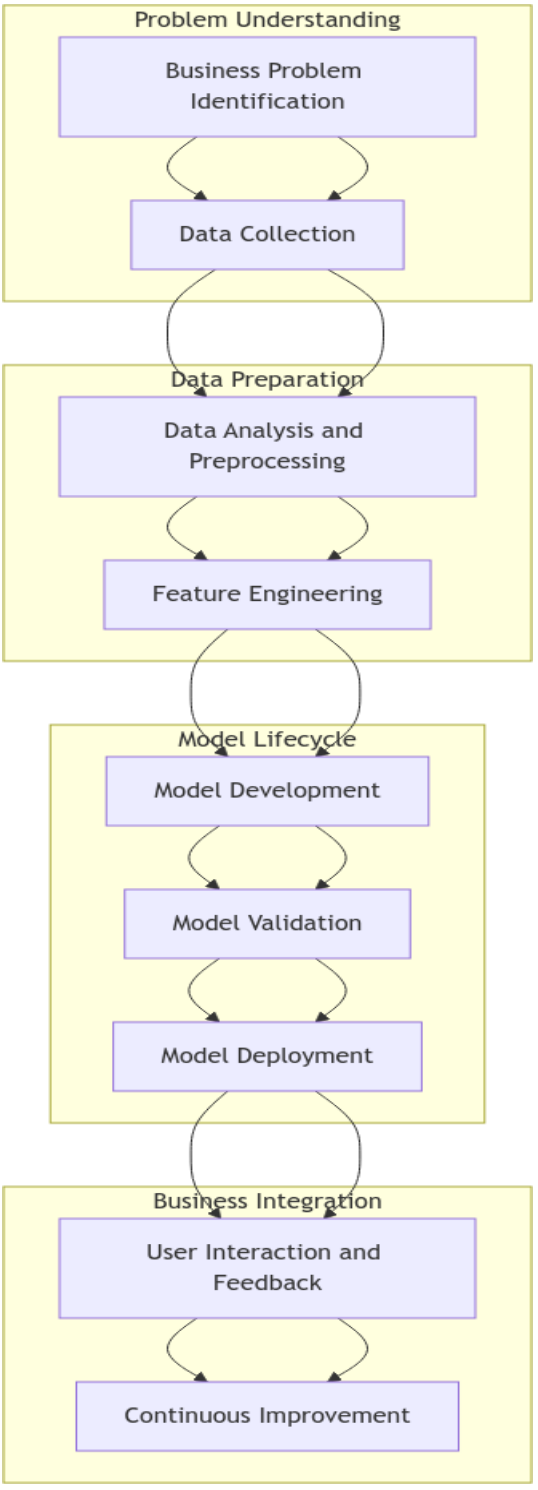


*Fig 3 Business diagram*

# 6. SYSTEM IMPLEMENTATION

## 6.1 Module description

### 1. Data Ingestion Module

**• Purpose and Role**

This module collects raw data from various sources like CSV files, databases, or APIs. It is responsible for automating the process of data retrieval and ensuring that the data pipeline remains robust and efficient.

**• Architecture**

The module connects to APIs or databases via secure protocols and integrates data into a staging area for preprocessing.

**• Key Features**

o Automated scheduling using cron jobs or task schedulers.

o Support for diverse file formats (CSV, JSON, SQL).

o Real-time data integration with APIs.

**• Implementation**

Frameworks like Python's pandas may be used.

**• Challenges**

Handling incomplete data or intermittent connectivity issues.

**• Real-World Applications**

Use of cloud services like AWS S3 for storing large-scale datasets.

### 2. Data Preprocessing Module

**• Purpose and Role**

This module cleans, transforms, and organizes the data to make it usable for modeling.

**• Steps Involved**

o Handling missing values using techniques like imputation.

o Encoding categorical variables (e.g., one-hot encoding).

o Normalizing numerical values for consistent scaling.

• **Key Features**

o Removal of outliers using statistical methods.

o Creation of derived features like sales growth rates or customer segmentation.

o Dimensionality reduction techniques (e.g., PCA).

• **Implementation**

Libraries such as scikit-learn and NumPy are critical for this phase.

• **Challenges**

Addressing class imbalances or ensuring the generalization of features across datasets.

• **Real-World Applications**

This module ensures data consistency across training and testing pipelines.

**3. Modeling Module**

• **Purpose and Role**

This module houses machine learning models that predict retail sales trends based on the preprocessed data.

• **Key Steps**

o Splitting data into training and validation sets.

o Training models like Random Forest, XGBoost

o Tuning hyperparameters using grid or random search methods.

• **Features**

o Model selection based on performance metrics like RMSE or R2.

o Comparison of multiple algorithms for optimal results.

o Logging of model training statistics for reproducibility.

• **Implementation**

Python libraries such as scikit-learn are used.

• **Challenges**

Overfitting, underfitting, or computational resource constraints.

• **Applications**

Forecasting sales by analyzing patterns in historical data.

**4. Evaluation Module**

**• Purpose and Role**

The module evaluates model performance using standard metrics.

**• Metrics Used**

- o Mean Absolute Error (MAE).
- o Root Mean Squared Error (RMSE).
- o R2 Score for regression tasks.

**• Features**

- o Generation of visualizations for performance comparison.
- o Cross-validation techniques to test robustness.
- o Error analysis for identifying model weaknesses.

**• Implementation**

Use libraries like matplotlib, seaborn, and scikit-learn.

**• Challenges**

Ensuring evaluation metrics align with project goals.

**• Applications**

Identifying the best model for deployment.

**5. Deployment Module**

**• Purpose and Role**

This module deploys the trained machine learning model to an accessible environment for real-time usage.

**• Key Steps**

- o Wrapping the model with APIs using Flask or FastAPI.
- o Hosting on platforms like AWS Lambda or Google Cloud Functions.
- o Ensuring low-latency and high-availability services.

**• Features**

- o Continuous Integration/Continuous Deployment (CI/CD) pipelines.
- o Docker containers for portability.
- o Secure access and authentication protocols.

• **Challenges**

Scaling the deployment for a global audience.

• **Applications**

Real-time sales predictions integrated with dashboards.

**6. User Interface (UI) Module**

• **Purpose and Role**

This module provides an interactive interface for end-users to access predictions and visual insights.

• **Key Features**

o Dashboards showing trends, sales predictions, and analytics.

o Interactive controls for filtering data or customizing predictions.

o Responsiveness for compatibility with mobile and desktop devices.

• **Implementation**

Tools like Streamlit, Dash, or web frameworks like React and Angular.

• **Challenges**

Balancing user experience with performance efficiency.

• **Applications**

Visual representation of key metrics and predictions.

**7. Logging and Monitoring Module**

• **Purpose and Role:**

Tracks the system's performance and logs errors for debugging and maintenance.

• **Features**

o Real-time monitoring of model and system health.

o Alerts for anomalies or system failures.

o Comprehensive logs for audit trails.

• **Implementation**

Tools like Prometheus, Grafana, or AWS CloudWatch.

• **Challenges**

Managing large volumes of logs efficiently.

**• Applications**

Ensuring reliability and availability of the deployed system.

**8. Feedback and Optimization Module**

**• Purpose and Role**

Collects feedback from users and improves model performance.

**• Key Features**

o Continuous learning mechanisms for updating models.

o A/B testing to refine predictions.

o Integration of user feedback loops into the training pipeline.

**• Implementation:**

Online learning techniques and reinforcement learning models.

**• Challenges:**

Handling noisy feedback or conflicting user inputs.

**• Applications:**

Improving system accuracy over time

### 6.1.1 Libraries used

**1. Pandas**

Pandas is one of the most widely used and powerful open-source data analysis and manipulation libraries in Python. Designed to handle structured data efficiently, it is particularly popular in data science, machine learning, and other data-intensive fields. The library's name, "Pandas," originates from the term *Panel Data*, which refers to multi- dimensional data sets. Since its introduction in 2008 by Wes McKinney, Pandas has become an indispensable tool for data professionals and researchers worldwide, owing to its simplicity, flexibility, and functionality.

**Key Features of Pandas**

1. **Data Structures**:
    - **Series**: A one-dimensional labeled array capable of holding data of any type (integer, string, float, etc.). Each element in a Series has an

associated label, known as an index, which makes data retrieval intuitive and user-friendly.

- **Data Frame**: A two-dimensional, tabular data structure similar to an Excel spreadsheet or a SQL table. A Data Frame consists of rows and columns, each of which can hold data of different types. The labeled indices and column names make it highly suitable for organizing and exploring complex datasets.

2. **Data Manipulation**: Pandas provides robust tools for reshaping and transforming data. Tasks such as filtering rows, selecting specific columns, grouping data, and applying functions can be performed effortlessly. Whether you're merging multiple datasets or pivoting data for analysis, Pandas offers built-in functions to streamline these processes.
3. **Data Cleaning**: Handling messy or incomplete data is a common challenge in real- world applications. Pandas simplifies tasks such as filling missing values, dropping duplicates, detecting outliers, and handling incorrect data types. Its ability to handle time-series data, missing timestamps, and other irregularities makes it particularly valuable for financial and scientific analyses.
4. **Indexing and Slicing**: The indexing capabilities of Pandas are unparalleled, allowing users to access and modify specific portions of a dataset quickly. Whether you want to slice rows using labels, retrieve specific values based on conditions, or work with hierarchical data, Pandas provides intuitive options for navigating data.
5. **File I/O**: Pandas can read from and write to various file formats, including CSV, Excel, JSON, SQL databases, and even Big Data frameworks like Apache Parquet. This interoperability makes it a versatile choice for integrating with other tools and systems.
6. **Time-Series Data**: One of Pandas' standout features is its support for time-series data. It includes functions for resampling, rolling-window computations, and time-zone handling, which are essential for financial analysis, stock trading, and other time- sensitive domains.

7. **Integration with Other Libraries**: Pandas seamlessly integrates with other Python libraries, such as Matplotlib, Seaborn, and Scikit-learn. This allows users to combine data manipulation with visualization and machine learning workflows effortlessly.

**Purpose**:
Pandas is an open-source library that provides high-performance data manipulation and analysis tools. It is particularly useful for working with structured data such as tables.

**Key Features**:

- Data Frames: Tabular data structures with labeled axes (rows and columns).
- Data cleaning: Handle missing data, duplicates, and outliers.
- Data merging and reshaping: Combine datasets or reorganize data for analysis.
- Grouping and aggregation: Perform summary statistics and complex transformations.

**Use in Project**:

Pandas was used to load the dataset, clean and preprocess data, and generate features for model training.

**NumPy**

NumPy (Numerical Python) is a powerful open-source Python library for numerical computing. It provides efficient tools for working with large, multi-dimensional arrays and matrices, along with a wide range of mathematical functions. Created by Travis Oliphant in 2006, NumPy is a core component of the Python scientific ecosystem.

At its heart is the ndarray (n-dimensional array), a fast, memory-efficient structure that supports vectorized operations—enabling element-wise computations without explicit loops. NumPy also offers advanced features like broadcasting, array reshaping, and indexing. With support for linear algebra, statistics, and random number generation, and seamless integration with libraries like Pandas, Matplotlib, and SciPy, NumPy is essential for data science, machine learning, and scientific computing. Efficient, reliable, and scalable, NumPy remains a fundamental tool for anyone working with data in Python.

**Purpose**: NumPy (Numerical Python) is a library for numerical computation in Python. It provides support for multi-dimensional arrays and matrices and includes a collection of mathematical functions to operate on these structures.

**Key Features**:

- Arrays: Efficient handling of large datasets.
- Mathematical operations: Support for linear algebra, statistics, and random number generation.
- Broadcasting: Element-wise operations on arrays with different shapes.

**Use in Project**: NumPy was used for numerical computations, including matrix operations and generating random data for testing models.

### 2. Matplotlib

Matplotlib is a popular open-source Python library for creating static, interactive, and dynamic visualizations. Developed by John D. Hunter in 2003, it supports a wide range of plot types, including line charts, bar graphs, scatter plots, and heatmaps.

Its pyplot module offers an easy-to-use, MATLAB-like interface ideal for beginners, while an object-oriented API caters to advanced users seeking detailed control. Matplotlib allows full customization of plots—titles, labels, colors, axes, and more—and integrates seamlessly with libraries like NumPy, Pandas, and Jupyter Notebooks.

Though it has a learning curve for complex plots, its flexibility, active community, and wide adoption make it a core tool in Python's data visualization ecosystem. Whether for simple data summaries or publication-quality graphics, Matplotlib remains a top choice.

**Purpose**: Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python.

**Key Features**:

- Plotting: Line plots, bar plots, scatter plots, and more.
- Customization: Control over figure size, labels, legends, and colors.

- Integration: Works seamlessly with Pandas and NumPy.

**Use in Project**: Matplotlib was used to create visualizations for exploratory data analysis, such as trends, distributions, and comparisons.

### 3. Seaborn

Seaborn is a Python data visualization library built on top of Matplotlib, designed to make creating informative and aesthetically pleasing statistical graphics easy and intuitive. Since its introduction by Michael Waskom, Seaborn has become one of the most popular libraries for visualizing data, especially among data scientists and analysts. Its primary strength lies in its ability to create high-level, statistical plots with minimal code, enabling users to focus on interpreting insights rather than struggling with complex customization. Seaborn is particularly well-suited for exploring relationships between variables, identifying trends, and analyzing patterns in datasets.

**Core Features of Seaborn**

1. **Simplified Syntax**: Seaborn abstracts away much of the complexity associated with Matplotlib, allowing users to generate professional-quality visualizations with a few lines of code. Its functions are tailored for working with structured datasets, such as Pandas DataFrames, making it a natural companion for data manipulation and analysis tasks.

2. **Statistical Graphics**: Seaborn is explicitly designed to handle statistical data. It provides built-in functions for creating common statistical plots, such as scatter plots, histograms, bar charts, and boxplots, along with more advanced options like violin plots, pair plots, and regression plots. These visuals are essential for understanding distributions, relationships, and group-level comparisons in data.

3. **Integration with Pandas**: One of Seaborn's standout features is its seamless integration with Pandas. Users can directly pass Pandas DataFrames or Series to Seaborn functions, which automatically map variables to plot elements. This integration allows for effortless visualization of real-world datasets without the need for extensive preprocessing.

4. **Automatic Aesthetics**: Seaborn includes aesthetically pleasing themes by default, ensuring that plots are visually appealing without requiring manual adjustments. It handles elements like color palettes, gridlines, and axis labels gracefully, creating clean and polished visuals.

5. **Customizability**: While Seaborn simplifies plotting, it retains flexibility for advanced users. It integrates with Matplotlib, allowing users to customize every aspect of their plots, from figure size to font style. This dual capability makes Seaborn accessible to beginners while providing advanced features for experienced users.

6. **Rich Variety of Plots**: Seaborn supports a wide range of plot types:

    - **Relational plots**: Scatter and line plots for analyzing relationships between variables.
    - **Distribution plots**: Histograms, KDE plots, and rug plots for visualizing data distributions.
    - **Categorical plots**: Bar plots, boxplots, and violin plots for exploring categorical data.
    - **Matrix plots**: Heatmaps and cluster maps for visualizing correlations or hierarchical relationships.
    - **Pair plots**: Multi-variable scatter plots for quickly exploring pairwise relationships in datasets.

7. **Color Palettes**: Seaborn provides a variety of built-in color palettes that are not only visually appealing but also designed for effective communication of data.

    These palettes include options for qualitative, sequential, and diverging data, enabling users to tailor their visualizations to specific needs.

8. **Statistical Estimations**: Seaborn simplifies common statistical operations, such as computing means, medians, confidence intervals, and regression lines.Many plotting functions include built-in options to perform these calculations, reducing the need for separate statistical analysis.

9. **Theme Customization**: Seaborn allows users to switch between themes (e.g., "darkgrid," "whitegrid," or "ticks") with a single command. This makes it easy to adapt plots to different presentation styles or preferences.

10. **Facet Grids**: A powerful feature of Seaborn is its ability to create facet grids, which allow users to visualize relationships across subsets of data. For example, users can generate a grid of plots showing trends for different categories or time periods, providing deeper insights into complex datasets.

**Purpose**: Seaborn is a Python visualization library built on top of Matplotlib. It provides a high-level interface for creating attractive and informative statistical graphics.

**Key Features**:

- Built-in themes: Enhance visual appeal.
- Statistical plots: Heatmaps, pair plots, and violin plots.
- Integration: Easily works with Pandas DataFrames.

**Use in Project**: Seaborn was used for creating advanced visualizations like heatmaps and pair plots to understand relationships between variables.

#### 4. Scikit-learn

Scikit-learn is a widely-used, open-source Python library that simplifies the implementation of machine learning algorithms. Built on top of foundational libraries like NumPy, SciPy, and Matplotlib, it provides efficient tools for data preprocessing, modeling, evaluation, and selection. With its intuitive and consistent API, Scikit-learn is accessible to both beginners and experienced users.

The library supports a wide variety of supervised and unsupervised learning

algorithms. For supervised learning, it includes popular models like Decision Trees, Random Forests, k-Nearest Neighbors, Support Vector Machines, and Logistic Regression. In unsupervised learning, it supports techniques such as K-Means Clustering, DBSCAN, and Principal Component Analysis (PCA).

Scikit-learn also offers robust model evaluation tools, including metrics like accuracy, precision, recall, F1-score, and confusion matrices. Its built-in methods for train-test splitting, cross-validation, and hyperparameter tuning (e.g., GridSearchCV) help users build reliable and well-generalized models.

In terms of data preprocessing, Scikit-learn includes utilities for handling missing values, scaling features, encoding categorical variables, and selecting relevant features. It also supports ensemble methods like Random Forest and Gradient Boosting, which enhance performance by combining multiple models.

While it does not support deep learning or big data out-of-the-box, Scikit-learn is ideal for traditional machine learning tasks and medium-sized datasets. Its versatility and ease of integration with other Python tools make it suitable for applications across industries such as finance, healthcare, marketing, and more.

In summary, Scikit-learn is a comprehensive and efficient library for traditional machine learning, offering the tools necessary to handle the full model development pipeline in a user-friendly and consistent way.

**Purpose**: Scikit-learn is a machine learning library in Python that provides simple and efficient tools for data mining and analysis.

**Key Features**:

- Supervised and unsupervised learning: Includes regression, classification, clustering, and dimensionality reduction algorithms.
- Preprocessing: Tools for feature scaling, encoding, and splitting data.
- Model evaluation: Metrics for assessing model performance.

**Use in Project**: Scikit-learn was used for splitting data, training baseline models, and evaluating performance.

- **Accuracy**

In machine learning, **accuracy** is a metric used to evaluate the performance of a model, particularly in classification tasks. It is defined as the ratio of the number of correct predictions made by the model to the total number of predictions. In other words, it measures the proportion of instances in a dataset that the model has correctly classified.

Mathematically, accuracy is expressed as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

Where:

5.3.1 **Correct Predictions** are the instances where the model's predicted label matches the true label (i.e., the model's output is correct).

5.3.2 **Total Predictions** refer to the total number of instances or samples the model has made predictions for.

For example, suppose you have a dataset with 100 samples, and your model correctly predicts 85 of them. The accuracy of the model would be:

$$\text{Accuracy} = \frac{85}{100} = 0.85 \quad \text{or} \quad 85\%$$

**For this project**

```
# Model Performance:
* **Model Accuracy**:
  * Random Forest              :     **0.99**
  * Gradient Boosting          :     **0.99**
  * Decision Tree              :     **0.99**
  * Bagging Regressor          :     **0.99**
  * Extra Tree Regressor       :     **0.99**
  * XGBoost                    :     **0.99**
  * Adaptive Boosting          :     **0.92**
  * K-Nearest Neighbour        :     **0.90**
  * Linear Regression          :     **0.82**
  * Support Vector Machine     :     **0.60**
```

*Fig 6.1.1 Accuracy*

## 6.2 VALIDATION CHECKS

To ensure the robustness and reliability of the machine learning models used in this project, several validation techniques were implemented. These checks help assess the model's performance, detect overfitting, and confirm its ability to generalize to unseen data. The following validation strategies were applied:

**1. Train-Test Split**

The dataset was divided into two primary subsets: a training set and a test set. Typically, 70–80% of the data was allocated for training the model, and the remaining 20–30% was reserved for testing. This helps evaluate how well the model performs on new, unseen data.

**2. Cross-Validation**

To further validate the model's consistency, k-fold cross-validation was used. In this technique, the training data is split into k equal parts (commonly 5 or 10). The model is trained on k-1 parts and tested on the remaining part. This process is repeated k times, and the results are averaged to provide a more reliable performance estimate.

**3. Performance Metrics**

Several performance metrics were used based on the problem type (classification or regression)

Accuracy, Precision, Recall, and F1-score for classification tasks.

Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and $R^2$ score for regression tasks. These metrics help assess not just overall performance, but also how well the model handles specific outcomes (e.g., false positives or variance).

**4. Confusion Matrix**

For classification problems, a confusion matrix was used to visualize and analyze model performance across different classes. This matrix helps in understanding the types of errors the model is making and where improvements may be needed.

**5. Overfitting and Underfitting Analysis**

Model behavior was monitored to detect signs of overfitting (high accuracy on training data but low on test data) or underfitting (poor performance on both training and test data). Regularization techniques and model complexity adjustments were used as needed.

**6. Hyperparameter Tuning**

Grid Search and Random Search methods were employed to find the optimal set of hyperparameters. This ensures the model is fine-tuned to deliver the best possible performance without overfitting.

**7. Data Preprocessing Validation**

Data preprocessing steps such as scaling, encoding, and handling missing values were validated to ensure consistency between training and testing phases. Pipelines were used to maintain preprocessing integrity during cross-validation and inference.

**8. Model Comparison**

Multiple models were trained and validated to compare their performance. The model with the best trade-off between bias and variance, and the highest cross-validation score, was selected for final deployment.

# 7. TESTING

## 7.1 Test cases

| TEST CASES | TEST CASE DESCRIPTION | TEST PROCEDURE | TEST DATA | EXPECTED RESULT | ACTUAL RESULT | STATUS |
|---|---|---|---|---|---|---|
| TC_F01 | Check UI Launch | Run streamlit run app_new.py | N/A | UI loads with all input fields | UI launches successfully | Pass |
| TC_F02 | Input Field Presence | Verify all input fields are visible | All text fields for features | All fields are rendered properly | All fields displayed correctly | Pass |
| TC_F03 | Model Load Check | Ensure final_model.pkl loads correctly | Valid model file | Model loaded without error | Model loaded successfully | Pass |
| TC_F04 | Column Mapping | Check if column_names.pkl loads correctly | Valid column name list | Input fields match expected features | Columns loaded correctly | Pass |
| TC_F05 | Valid Input Prediction | Enter valid inputs and click "Predict" | Sample: Electronics, Male, 30, etc. | Model predicts and returns a sales value | Prediction displayed with result | Pass |
| TC_F06 | Empty Input Handling | Submit form with some empty fields | Leave a few fields blank | App should warn or show controlled error | Error message shown or handled gracefully | Pass |

| | | | | | | |
|---|---|---|---|---|---|---|
| TC_F 07 | Numeric Conversion | Test number fields with valid inputs | Example: Quantity = 3, Sales = 200.50 | Input converted and passed correctly | Predictio n computed as expected | Pass |
| TC_F 08 | Output Formatting | Ensure predicted result is rounded | Any valid input | Output in two decimal format | Example: "The Sales is 1573.47" | Pass |

# 8.CONCLUSION AND FUTURE SCOPE

## 8.1 Conclusion

This research aimed to explore the potential of machine learning, particularly Random Forest algorithms, to predict global retail sales for a popular fictional influencer's online store. Using a dataset sourced from Kaggle that includes order details, product information, customer demographics, and sales metrics, the study integrated sentiment analysis and time series analysis to assess how these factors influence sales performance. The Random Forest model was chosen for its ability to handle large datasets, manage complex relationships between variables, and provide valuable insights in prediction accuracy.This research successfully demonstrated the application of machine learning, particularly the Random Forest algorithm, in predicting total sales for a global retail business inspired by a fictional influencer's online store. By leveraging a comprehensive dataset sourced from Kaggle, which included key dimensions such as order details, product information, customer demographics, and sales metrics, this study highlights the pivotal role of data in driving business strategies. Through detailed analysis, factors such as product category, customer demographics, and international shipping emerged as significant contributors to sales performance, offering a deeper understanding of the interplay between these variables.

The choice of the Random Forest model proved advantageous due to its robustness, interpretability, and efficiency in handling complex datasets with high-dimensional features. The model not only predicted sales with notable accuracy but also provided insights into feature importance, enabling businesses to identify key drivers of success. For example, customer demographics like age, income, and location were critical in shaping purchase behaviors, while product-specific attributes like category and price influenced buying decisions. Moreover, the analysis of international shipping data shed light on the logistical aspects that can impact customer satisfaction and repeat purchases, emphasizing the importance of seamless global operations in today’s interconnected marketplace.

The integration of sentiment analysis and time series analysis added another layer of depth to the study. Sentiment analysis helped identify trends in customer satisfaction

and preferences by analyzing customer reviews and feedback, while time series analysis provided a forward- looking perspective on sales trends. These techniques allowed for a holistic approach, combining historical data with predictive insights to guide future strategies. By identifying seasonal patterns, peak sales periods, and potential demand fluctuations, businesses can optimize their inventory, marketing campaigns, and pricing strategies.

Streamlit was instrumental in translating complex analytical results into an accessible and interactive format. The platform allowed stakeholders to interact with the data in real time, exploring trends and understanding the model's predictions through intuitive visualizations. This accessibility fosters collaboration and ensures that data-driven insights are actionable, empowering decision-makers to implement strategies based on evidence rather than intuition. The use of Streamlit not only enhances the presentation of analytical findings but also bridges the gap between data scientists and business leaders, creating a seamless flow of information.

From a business perspective, this research underscores the importance of adopting advanced analytical techniques to stay competitive in the e-commerce space. Predictive modeling provides a strategic advantage, enabling businesses to anticipate market trends, understand customer behavior, and tailor their offerings to meet evolving demands. The findings suggest that investing in data-driven solutions can result in significant improvements in sales performance, customer satisfaction, and operational efficiency. Furthermore, the insights gained from this study highlight the value of personalization and customer-centric approaches in fostering loyalty and driving growth.Looking ahead, this study opens avenues for further exploration. Expanding the dataset to include additional variables, such as marketing spend, regional cultural preferences, or external economic factors, could refine the predictive model and provide even more nuanced insights. Incorporating advanced machine learning techniques like neural networks or ensemble methods might improve accuracy and scalability. Additionally, exploring the impact of emerging trends, such as sustainable practices or social media influences, on sales performance could provide valuable insights for modern retail strategies.

In conclusion, this research demonstrates the transformative potential of machine learning and interactive data visualization in modern business analytics. By leveraging tools like Random Forest and Streamlit, businesses can harness the power of data to optimize their operations and adapt to the dynamic e-commerce landscape. As the global retail environment continues to evolve, data-driven decision-making will remain a cornerstone of success, enabling businesses to stay ahead of the competition and deliver exceptional value to their customers.

## 8.2 Future Scope

### 1. Advancements in Predictive Analytics for Retail

- **Emerging Technologies:**
  Discuss how advanced machine learning techniques like reinforcement learning and deep learning architectures could further enhance the accuracy of retail sales predictions.
    - **Integration with Big Data:** The increasing availability of large-scale, real-time data from IoT devices, social media, and customer behavior tracking systems could significantly enhance prediction capabilities.
    - **AI-driven Personalization:** Retailers can use AI models to create highly personalized customer experiences, improving retention and sales.
    - **Natural Language Processing (NLP):** NLP can be utilized for sentiment analysis and demand forecasting by analyzing customer feedback and market trends.
- **Scalability in Cloud Computing:**
  With the evolution of cloud infrastructure, retail models can leverage scalable resources to handle large-scale global predictions seamlessly.

### 2. Enhanced Integration with Business Intelligence Tools

- **Real-time Insights:**
  Integration of predictive models with dynamic dashboards, such as Power

BI or Tableau, will allow businesses to monitor predictions in real time, aiding in better decision-making.

- **Predictive vs. Prescriptive Analytics:** Moving from merely predicting outcomes to prescribing actionable strategies (e.g., optimal pricing, inventory adjustments).
- **Scenario Analysis:** Simulation capabilities could allow businesses to analyze "what-if" scenarios effectively.

- **Automated Decision-making Systems:**
  Developments in AI-driven systems could enable automatic adjustment of pricing strategies, inventory orders, and marketing campaigns based on predictive analytics.

**3. Evolution of Retail Trends and Adaptation**

- **Omnichannel Retailing:**
  Future predictive systems will likely focus on seamless customer experiences across multiple channels, including e-commerce, brick-and-mortar, and social commerce.
  - **Consumer Behavior Analysis:** Deeper analysis of trends such as sustainability preferences, product return patterns, and seasonal demands.
  - **Hyper-local Insights:** Retailers can adapt predictions to local demand patterns, helping businesses cater to specific community needs.
- **Global Expansion:**
  As global markets evolve, predictive models will need to incorporate diverse datasets across regions, accounting for cultural, economic, and seasonal variances.

**4. Sustainable and Ethical Practices in AI**

- **Eco-Friendly Retail Predictions:**
  Predictive analytics can play a crucial role in minimizing waste by optimizing inventory levels and reducing overproduction.

- **Carbon Emission Reduction:** Retailers could predict and optimize logistics for reduced carbon footprints.
- **Ethical AI:** Implementing transparent and unbiased AI models to ensure ethical decision-making in areas like pricing and promotions.

- **Future Regulatory Challenges:**

  Discuss potential data privacy regulations and how evolving AI ethics could shape the development and deployment of predictive models in the retail space.

# 9.APPENDIX

## 9.1 Source code

```
import numpy as np
import pandas as pd
import datetime as dt
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv('/content/merch_sales.csv')
df.head()
df.shape
df.info()
df.describe()
df.describe().transpose()
df.columns
df.select_dtypes(include='object').columns

# get all numerical columns
df.select_dtypes(include='number').columns
df.isnull().sum()

# check for missing values
df.isnull().sum().any()

df = df.drop(['Order ID','Product ID','Review'],axis=1)
df.head()
df['Year'] = pd.DatetimeIndex(df['Order Date']).year
df['Month'] = pd.DatetimeIndex(df['Order Date']).month
df['Day'] = pd.DatetimeIndex(df['Order Date']).day
df.head()

# drop the Order_Date column
df = df.drop(['Order Date'],axis=1)
df.head()
df.shape
df.info()
df.describe()

df.describe().transpose()
df.columns
df.select_dtypes(include='object').columns
df.select_dtypes(include='number').columns

numerical = df.select_dtypes(exclude='object').columns
numerical
```

```
# get all the categorical columns
categorical = df.select_dtypes(include='object').columns
categorical
for column in numerical:
  plt.figure(figsize=(10,6),dpi=200)
  sns.histplot(df[column],color = 'green',edgecolor = 'black')
  plt.title(column)
  plt.xlabel(column)
  plt.show()

# plot a histogram for numerical columns along with kernel density estimator
for column in numerical:
  plt.figure(figsize=(10,6),dpi=200)
  sns.histplot(df[column],color = 'green',edgecolor = 'black',kde='True')
  plt.title(column)
  plt.xlabel(column)
  plt.show()

# plot a boxplot for categorical columns
for column in categorical:
  plt.figure(figsize=(10,6),dpi=200)
  sns.boxplot(df[column])
  plt.title(column)

# plot a count plot for categorical columns
for column in categorical:
  plt.figure(figsize=(10,6),dpi=200)
  sns.countplot(data=df,x=df[column])
  plt.xticks(rotation=90)
  plt.title(column)

df.columns

# get all the features
features = df.drop(['Total Sales'],axis=1)
features.columns

# plot a scatterplot along with hue
for column in features:
  plt.figure(figsize=(10,6),dpi=200)
  sns.scatterplot(data=df,x=df[column],y=df['Total Sales'],hue='International Shipping')
  plt.xticks(rotation=90)
  plt.title(column)

# plot a histogram for numerical columns along with hue
for column in numerical:
  plt.figure(figsize=(10,6),dpi=200)
  sns.histplot(data=df, x=column, hue='International Shipping')
```

```
  plt.title(column)
  plt.xlabel(column)
  plt.show()

# plot a histogram for numerical columns along with hue and kernel density estimator
for column in numerical:
  plt.figure(figsize=(10,6),dpi=200)
  sns.histplot(data=df, x=column, color='green', edgecolor='black', hue='International
Shipping',kde=True)
  plt.title(column)
  plt.xlabel(column)
  plt.show()

# plot a count plot for categorical columns along with hue
for column in categorical:
  plt.figure(figsize=(10,6),dpi=200)
  sns.countplot(data=df,x=df[column],hue='International Shipping')
  plt.xticks(rotation=90)
  plt.title(column)
  plt.show()

# plot a violin plot for categorical columns along with hue
for column in categorical:
  plt.figure(figsize=(10,6),dpi=200)
  sns.violinplot(data=df,x=df[column],hue='International Shipping')
  plt.xticks(rotation=90)
  plt.title(column)
  plt.show()

df.corr(numeric_only=True)

# plot a heatmap
plt.figure(figsize=(10,6),dpi=200)
sns.heatmap(df.corr(numeric_only=True),annot=True)
plt.show()

df['Product Category'].unique()

# get all unique values from Buyer Gender
df['Buyer Gender'].unique()

# get all unique values from International Shipping
df['International Shipping'].unique()

# get all unique values from Order Location
df['Order Location'].unique()
df['Order Location'].nunique()
```

```
from sklearn.preprocessing import LabelEncoder

# create an instance for LabelEncoder()
encoder = LabelEncoder()

df['Product Category'] = encoder.fit_transform(df['Product Category'])
df['Buyer Gender'] = encoder.fit_transform(df['Buyer Gender'])
df['International Shipping'] = encoder.fit_transform(df['International Shipping'])
df['Order Location'] = encoder.fit_transform(df['Order Location'])
df.head()
X = df.drop(['Total Sales'],axis=1)
y = df['Total Sales']
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# get the number of rows and columns in X_train
X_train.shape

# get the number of rows and columns in X_test
X_test.shape

# get the number of rows and columns in y_train
y_train.shape

# get the number of rows and columns in y_test
y_test.shape

### Validation

# holdout
X_validation, X_holdout_test, y_validation, y_holdout_test = train_test_split(X_test,
y_test, test_size=0.5, random_state=42)

# get the number of rows and columns in X_validate
X_validation.shape

# get the number of rows and columns in X_holdout_test
X_holdout_test.shape

# get the number of rows and columns in X_validate
y_validation.shape

# get the number of rows and columns in y_holdout_test
y_holdout_test.shape

from sklearn.preprocessing import StandardScaler
# create an Instance for StandardScaler
scaler = StandardScaler()
```

```
# scale the training data
scaled_X_train = scaler.fit_transform(X_train)
# transform the test data
scaled_X_test = scaler.transform(X_test)
# scale the validation data
scaled_X_validation = scaler.transform(X_validation)
# scale the holdout data
scaled_X_holdout_test = scaler.transform(X_holdout_test)
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score

# create an Instance of the model
lr_model = LinearRegression()
# fit the model
lr_model.fit(scaled_X_train, y_train)
# predict the scaled_X_validation
y_val_pred_lr = lr_model.predict(scaled_X_validation)
# predict the scaled_X_holdout_test
y_pred_lr = lr_model.predict(scaled_X_holdout_test)

r2_score(y_validation, y_val_pred_lr)

# model evaluation for validation data using mean_absolute_error
mean_absolute_error(y_validation, y_val_pred_lr)

# model evaluation for validation data using mean_squared_error
mean_squared_error(y_validation, y_val_pred_lr)

# model evaluation for validation data using root mean squared error
np.sqrt(mean_squared_error(y_validation, y_val_pred_lr))

# model evaluation for validation data using median_absolute_error
median_absolute_error(y_validation, y_val_pred_lr)

# model evaluation for validation data using explained_variance_score
explained_variance_score(y_validation, y_val_pred_lr)

r2_score(y_holdout_test, y_pred_lr)

# model evaluation using mean_absolute_error
mean_absolute_error(y_holdout_test, y_pred_lr)

# model evaluation using mean_squared_error
mean_squared_error(y_holdout_test, y_pred_lr)

# model evaluation using root mean squared error
np.sqrt(mean_squared_error(y_holdout_test, y_pred_lr))
```

```
# model evaluation using median_absolute_error
median_absolute_error(y_holdout_test, y_pred_lr)

# model evaluation using explained_variance_score
explained_variance_score(y_holdout_test, y_pred_lr)
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score

# create an Instance of the model
rf_model = RandomForestRegressor()
# fit the model
rf_model.fit(scaled_X_train, y_train)
# predict the scaled_X_validation
y_val_pred_rf = rf_model.predict(scaled_X_validation)
# predict the scaled_X_holdout_test
y_pred_rf = rf_model.predict(scaled_X_holdout_test)
r2_score(y_validation, y_val_pred_rf)

# model evaluation for validation data using mean_absolute_error
mean_absolute_error(y_validation, y_val_pred_rf)

# model evaluation for validation data using mean_squared_error
mean_squared_error(y_validation, y_val_pred_rf)
np.sqrt(mean_squared_error(y_validation, y_val_pred_rf))

median_absolute_error(y_validation, y_val_pred_rf)

explained_variance_score(y_validation, y_val_pred_rf)

r2_score(y_holdout_test, y_pred_rf)

mean_absolute_error(y_holdout_test, y_pred_rf)

mean_squared_error(y_holdout_test, y_pred_rf)

np.sqrt(mean_squared_error(y_holdout_test, y_pred_rf))

median_absolute_error(y_holdout_test, y_pred_rf)

explained_variance_score(y_holdout_test, y_pred_rf)
from sklearn.svm import SVR
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score

# create an Instance of the model
svr_model = SVR()
```

```
# fit the model
svr_model.fit(scaled_X_train, y_train)
# predict the scaled_X_validation
y_val_pred_svr = svr_model.predict(scaled_X_validation)
# predict the scaled_X_holdout_test
y_pred_svr = svr_model.predict(scaled_X_holdout_test)
r2_score(y_validation, y_val_pred_svr)

# model evaluation for validation data using mean_absolute_error
mean_absolute_error(y_validation, y_val_pred_svr)

# model evaluation for validation data using mean_squared_error
mean_squared_error(y_validation, y_val_pred_svr)

# model evaluation for validation data using root mean squared error
np.sqrt(mean_squared_error(y_validation, y_val_pred_svr))

# model evaluation for validation data using median_absolute_error
median_absolute_error(y_validation, y_val_pred_svr)

# model evaluation for validation data using explained_variance_score
explained_variance_score(y_validation, y_val_pred_svr)
r2_score(y_holdout_test, y_pred_svr)

# model evaluation using mean_absolute_error
mean_absolute_error(y_holdout_test, y_pred_svr)

# model evaluation using mean_squared_error
mean_squared_error(y_holdout_test, y_pred_svr)

# model evaluation using root mean squared error
np.sqrt(mean_squared_error(y_holdout_test, y_pred_svr))

# model evaluation using median_absolute_error
median_absolute_error(y_holdout_test, y_pred_svr)

# model evaluation using explained_variance_score
explained_variance_score(y_holdout_test, y_pred_svr)

from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score

# create an Instance of the model
knn_model = KNeighborsRegressor()
# fit the model
knn_model.fit(scaled_X_train, y_train)
# predict the scaled_X_validation
```

```
y_val_pred_knn = knn_model.predict(scaled_X_validation)
# predict the scaled_X_holdout_test
y_pred_knn = knn_model.predict(scaled_X_holdout_test)
r2_score(y_validation, y_val_pred_knn)

# model evaluation for validation data using mean_absolute_error
mean_absolute_error(y_validation, y_val_pred_knn)

# model evaluation for validation data using mean_squared_error
mean_squared_error(y_validation, y_val_pred_knn)

# model evaluation for validation data using root mean squared error
np.sqrt(mean_squared_error(y_validation, y_val_pred_knn))

# model evaluation for validation data using median_absolute_error
median_absolute_error(y_validation, y_val_pred_knn)

# model evaluation for validation data using explained_variance_score
explained_variance_score(y_validation, y_val_pred_knn)
r2_score(y_holdout_test, y_pred_knn)

# model evaluation using mean_absolute_error
mean_absolute_error(y_holdout_test, y_pred_knn)

# model evaluation using mean_squared_error
mean_squared_error(y_holdout_test, y_pred_knn)

np.sqrt(mean_squared_error(y_holdout_test, y_pred_knn))

median_absolute_error(y_holdout_test, y_pred_knn)

explained_variance_score(y_holdout_test, y_pred_knn)

from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score

dt_model = DecisionTreeRegressor()
# fit the model
dt_model.fit(scaled_X_train, y_train)
# predict the scaled_X_validation
y_val_pred_dt = dt_model.predict(scaled_X_validation)
# predict the scaled_X_holdout_test
y_pred_dt = dt_model.predict(scaled_X_holdout_test)
r2_score(y_validation, y_val_pred_dt)

mean_absolute_error(y_validation, y_val_pred_dt)
mean_squared_error(y_validation, y_val_pred_dt)
```

```
np.sqrt(mean_squared_error(y_validation, y_val_pred_dt))

median_absolute_error(y_validation, y_val_pred_dt)
explained_variance_score(y_validation, y_val_pred_dt)
r2_score(y_holdout_test, y_pred_dt)

mean_absolute_error(y_holdout_test, y_pred_dt)
mean_squared_error(y_holdout_test, y_pred_dt)

np.sqrt(mean_squared_error(y_holdout_test, y_pred_dt))

median_absolute_error(y_holdout_test, y_pred_dt)
explained_variance_score(y_holdout_test, y_pred_dt)

from sklearn.ensemble import GradientBoostingRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score

gb_model = GradientBoostingRegressor()
# fit the model
gb_model.fit(scaled_X_train, y_train)
# predict the scaled_X_validation
y_val_pred_gb = gb_model.predict(scaled_X_validation)
y_pred_gb = gb_model.predict(scaled_X_holdout_test)
r2_score(y_validation, y_val_pred_gb)

mean_absolute_error(y_validation, y_val_pred_gb)

mean_squared_error(y_validation, y_val_pred_gb)

np.sqrt(mean_squared_error(y_validation, y_val_pred_gb))
median_absolute_error(y_validation, y_val_pred_gb)

explained_variance_score(y_validation, y_val_pred_gb)
r2_score(y_holdout_test, y_pred_gb)
mean_absolute_error(y_holdout_test, y_pred_gb)

mean_squared_error(y_holdout_test, y_pred_gb)

np.sqrt(mean_squared_error(y_holdout_test, y_pred_gb))

median_absolute_error(y_holdout_test, y_pred_gb)

explained_variance_score(y_holdout_test, y_pred_gb)
from sklearn.ensemble import AdaBoostRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score
```

```
ada_model = AdaBoostRegressor()
# fit the model
ada_model.fit(scaled_X_train, y_train)
# predict the scaled_X_validation
y_val_pred_ada = ada_model.predict(scaled_X_validation)
# predict the scaled_X_holdout_test
y_pred_ada = ada_model.predict(scaled_X_holdout_test)
r2_score(y_validation, y_val_pred_ada)
mean_absolute_error(y_validation, y_val_pred_ada)
mean_squared_error(y_validation, y_val_pred_ada)

np.sqrt(mean_squared_error(y_validation, y_val_pred_ada))

median_absolute_error(y_validation, y_val_pred_ada)

explained_variance_score(y_validation, y_val_pred_ada)
r2_score(y_holdout_test, y_pred_ada)

mean_absolute_error(y_holdout_test, y_pred_ada)

mean_squared_error(y_holdout_test, y_pred_ada)

np.sqrt(mean_squared_error(y_holdout_test, y_pred_ada))

median_absolute_error(y_holdout_test, y_pred_ada)
explained_variance_score(y_holdout_test, y_pred_ada)
from xgboost import XGBRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score

xgb_model = XGBRegressor()
xgb_model.fit(scaled_X_train, y_train)
y_val_pred_xgb = xgb_model.predict(scaled_X_validation)
y_pred_xgb = xgb_model.predict(scaled_X_holdout_test)

r2_score(y_validation, y_val_pred_xgb)

mean_absolute_error(y_validation, y_val_pred_xgb)

mean_squared_error(y_validation, y_val_pred_xgb)

np.sqrt(mean_squared_error(y_validation, y_val_pred_xgb))

median_absolute_error(y_validation, y_val_pred_xgb)
explained_variance_score(y_validation, y_val_pred_xgb)

r2_score(y_holdout_test, y_pred_xgb)
```

```
mean_absolute_error(y_holdout_test, y_pred_xgb)

mean_squared_error(y_holdout_test, y_pred_xgb)

np.sqrt(mean_squared_error(y_holdout_test, y_pred_xgb))

median_absolute_error(y_holdout_test, y_pred_xgb)
explained_variance_score(y_holdout_test, y_pred_xgb)
from sklearn.ensemble import BaggingRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score

bag_model = BaggingRegressor()
bag_model.fit(scaled_X_train, y_train)
# predict the scaled_X_validation
y_val_pred_bag = bag_model.predict(scaled_X_validation)
# predict the scaled_X_holdout_test
y_pred_bag = bag_model.predict(scaled_X_holdout_test)
r2_score(y_validation, y_val_pred_bag)

# model evaluation for validation data using mean_absolute_error
mean_absolute_error(y_validation, y_val_pred_bag)

mean_squared_error(y_validation, y_val_pred_bag)

np.sqrt(mean_squared_error(y_validation, y_val_pred_bag))

median_absolute_error(y_validation, y_val_pred_bag)

explained_variance_score(y_validation, y_val_pred_bag)
r2_score(y_holdout_test, y_pred_bag)

mean_absolute_error(y_holdout_test, y_pred_bag)

mean_squared_error(y_holdout_test, y_pred_bag)

np.sqrt(mean_squared_error(y_holdout_test, y_pred_bag))

median_absolute_error(y_holdout_test, y_pred_bag)

explained_variance_score(y_holdout_test, y_pred_bag)
from sklearn.ensemble import ExtraTreesRegressor
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error,
median_absolute_error, explained_variance_score

ext_model = ExtraTreesRegressor()
# fit the model
```

```
ext_model.fit(scaled_X_train, y_train)
# predict the scaled_X_validation
y_val_pred_ext = ext_model.predict(scaled_X_validation)
# predict the scaled_X_holdout_test
y_pred_ext = ext_model.predict(scaled_X_holdout_test)
r2_score(y_validation, y_val_pred_ext)
mean_absolute_error(y_validation, y_val_pred_ext)

mean_squared_error(y_validation, y_val_pred_ext)

np.sqrt(mean_squared_error(y_validation, y_val_pred_ext))

median_absolute_error(y_validation, y_val_pred_ext)
explained_variance_score(y_validation, y_val_pred_ext)

r2_score(y_holdout_test, y_pred_ext)
mean_absolute_error(y_holdout_test, y_pred_ext)
mean_squared_error(y_holdout_test, y_pred_ext)

np.sqrt(mean_squared_error(y_holdout_test, y_pred_ext))

median_absolute_error(y_holdout_test, y_pred_ext)

explained_variance_score(y_holdout_test, y_pred_ext)
import numpy as np
import pandas as pd
import datetime as dt
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('merch_sales.csv')
df.head()

df = df.drop(['Order ID','Product ID','Review'],axis=1)

df['Year'] = pd.DatetimeIndex(df['Order Date']).year
df['Month'] = pd.DatetimeIndex(df['Order Date']).month
df['Day'] = pd.DatetimeIndex(df['Order Date']).day
df = df.drop(['Order Date'],axis=1)
df.head()

from sklearn.preprocessing import LabelEncoder
encoder = LabelEncoder()

df['Product Category'] = encoder.fit_transform(df['Product Category'])
df['Buyer Gender'] = encoder.fit_transform(df['Buyer Gender'])
df['International Shipping'] = encoder.fit_transform(df['International Shipping'])
df['Order Location'] = encoder.fit_transform(df['Order Location'])
```

```
df.head()
X = df.drop(['Total Sales'],axis=1)
y = df['Total Sales']

X.head()

y.head()
from sklearn.preprocessing import StandardScaler
# create an Instance for StandardScaler
scaler = StandardScaler()
# scale the training data
scaled_X = scaler.fit_transform(X)
from sklearn.ensemble import RandomForestRegressor
# create an instance for the model
model = RandomForestRegressor()
# fitting the model
model.fit(scaled_X, y)
import joblib
joblib.dump(model,'final_model.pkl')
list(X.columns)

joblib.dump(list(X.columns),'column_names.pkl')
import joblib
import pandas as pd
from sklearn.preprocessing import StandardScaler
new_columns = joblib.load('column_names.pkl')
loaded_model = joblib.load('final_model.pkl')
data = [[0, 1, 30, 15, 0, 100, 0, 100, 1, 100, 4, 2024, 7, 21]]
input_df = pd.DataFrame(data, columns=new_columns + ['extra_column']) # adjust
column if needed
input_df = input_df[new_columns] # adjust column if needed
scaled_data = scaler.transform(input_df)
loaded_model.predict(scaled_data)

import joblib
new_columns = joblib.load('column_names.pkl')

new_columns
!pip install streamlit -q

%%writefile app_new.py
import pickle
import pandas as pd
import streamlit as st
import joblib
model = joblib.load('final_model.pkl')
new_columns = joblib.load('column_names.pkl')
```

```
st.title('Sales Prediction')

def main():
  st.title('Sales Prediction')
  Product_Category = st.text_input('Product Category')
  Buyer_Gender = st.text_input('Buyer Gender')
  Buyer_Age = st.text_input('Buyer Age')
  Order_Location = st.text_input('Order Location')
  International_Shipping = st.text_input('International Shipping')
  Sales_Price = st.text_input('Sales Price')
  Shipping_Charges = st.text_input('Shipping Charges')
  Sales_per_Unit = st.text_input('Sales per Unit')
  Quantity = st.text_input('Quantity')
  Rating = st.text_input('Rating')
  Year = st.text_input('Year')
  Month = st.text_input('Month')
  Day = st.text_input('Day')

  if st.button('Predict'):
    # Assuming your model expects data in the same order as new_columns
    input_data = [Product_Category, Buyer_Gender, Buyer_Age,Order_Location,
                  International_Shipping, Sales_Price,Shipping_Charges,
                  Sales_per_Unit, Quantity, Rating,Year, Month, Day]

    # Create a DataFrame for prediction
    input_df = pd.DataFrame([input_data], columns=new_columns)

    prediction = model.predict(input_df)
    output = round(prediction[0], 2)
    st.success('The Sales is {}'.format(output))

if __name__ == '__main__':
  main()

!wget -q -O - ipv4.icanhazip.com

!npm install -g localtunnel

!streamlit run app_new.py & npx localtunnel --port 8501
```

### 9.2 Screen shots


*Fig 9.2.1 Dashboard*

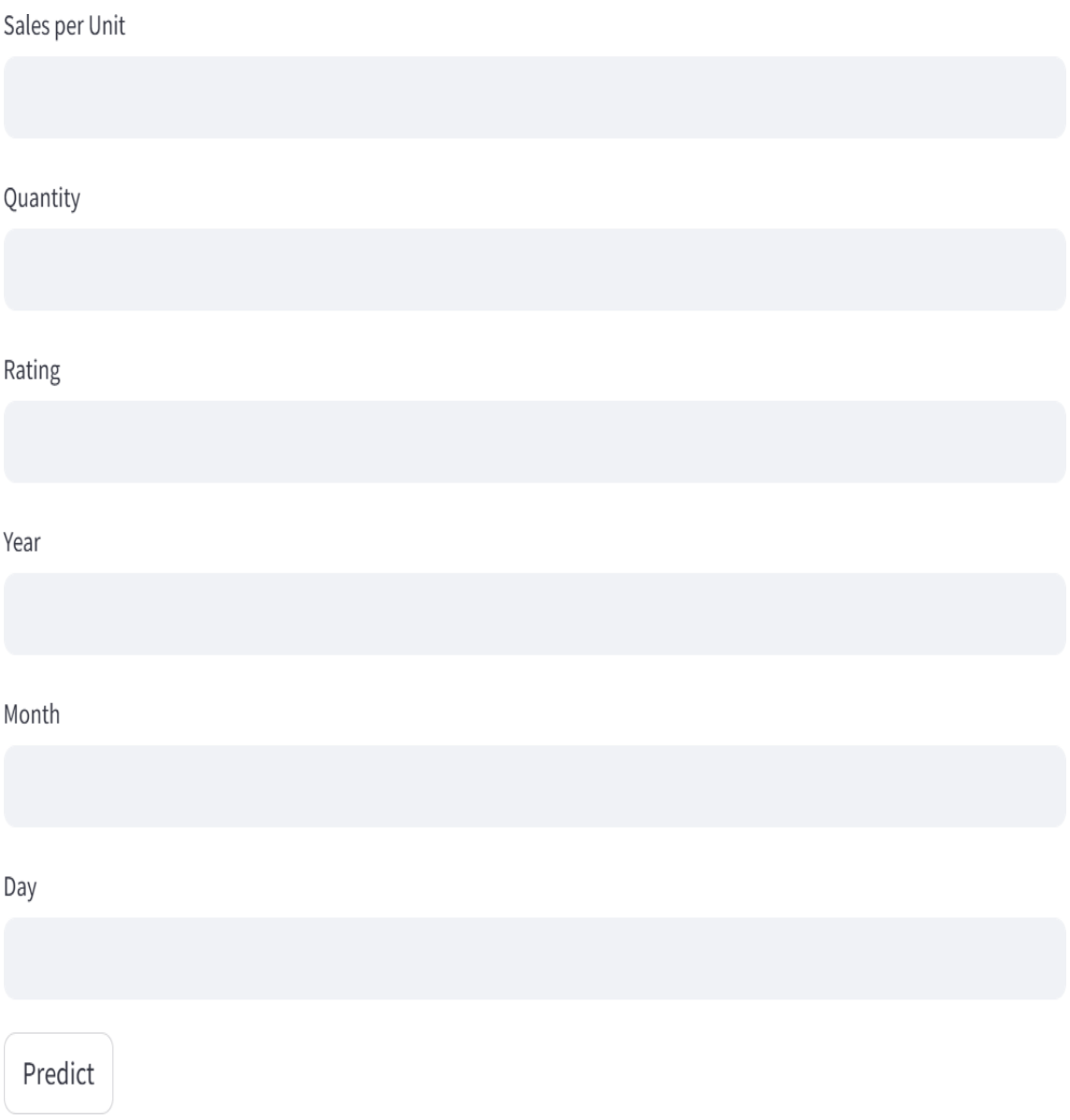


*Fig 9.2.2 Dashboard*

## 9.3 User documentation

This project focuses on predicting the retail sales price of products using historical retail transaction data. It incorporates machine learning techniques to model the relationship between various sales-related factors and the sales price. The final model is deployed using a user-friendly web interface built with Streamlit, allowing users to input values and receive real-time predictions.

**How to Use the Application**

**Step 1: Launch the Application**

1. Open the terminal or command prompt.
2. Navigate to the directory where the app_new.py file is located.
3. Run the following command:

   streamlit run app_new.py
4. The application will open in the default web browser at:

   http://localhost:8501

**Step 2: Enter Input Values**

Once the application loads, users will be prompted to enter the following input fields:

- Product Category
- Buyer Gender
- Buyer Age
- Order Location
- International Shipping (Yes/No)
- Sales Price
- Shipping Charges
- Sales per Unit
- Quantity
- Rating
- Year
- Month
- Day

These inputs are essential for the machine learning model to make accurate predictions.

**Step 3: Make a Prediction**

After filling all the required input fields, click the **"Predict"** button. The system will process the inputs and display the predicted sales price on the screen.

---

**Backend and Model Details**

- **Algorithm Used**: Random Forest Regressor
- **Libraries Used**: Pandas, Scikit-learn, Streamlit, Joblib
- **Model Input Format**: A feature vector that matches the training schema
- **Output**: Predicted sales price (float value, rounded to two decimal places)

## 9.4 Glossary

**Streamlit** refers to a Python-based framework used for building interactive web applications specifically for machine learning and data science use cases.

**Pickle files**, with the extension .pkl, are used to serialize and store Python objects. In this project, they are used to save the trained machine learning model and column configuration.

**Random Forest** is an ensemble machine learning algorithm that uses multiple decision trees to improve the accuracy and robustness of predictions, especially suitable for regression and classification tasks.

**Model Deployment** is the act of integrating a trained machine learning model into a usable application so that end users can input data and receive predictions in real time.

**Sales per Unit** represents the pricing of each individual item in a transaction and helps capture product-level pricing behavior.

**Joblib** is a Python library used for the efficient storage and retrieval of large data objects, particularly suited for machine learning models and numpy arrays.

**User Interface (UI)** is the part of the application through which users interact with the system. In this case, the UI is built using Streamlit and allows users to enter data and receive the output prediction.

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